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A human neuron Alzheimer's disease model reveals barriers to senolytic translatability

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Abstract

Therapeutic successes in mouse models of Alzheimer's disease (AD) largely fail to translate into clinical trials, with experimental drugs rarely validated in human models before being administered to humans. To address this, we developed an accessible method to culture commercially available primary human neurons and astrocytes, along with an amyloid-beta 1–42 (A β)-based in vitro AD model. This system enables to reliably culture primary neurons to mature stages of development, essential to model the adult brain and neurodegenerative diseases such as AD. The absence of a blood–brain barrier (BBB) in this model permits evaluation of drug mechanisms of action independently of BBB permeability, thereby informing the feasibility of developing fully BBB-penetrant therapeutics for CNS interventions based on validated senolytic pathways. Using this platform, we evaluated two senolytic regimens previously shown to be effective in AD mouse models: Navitoclax (NAV), which targets the Bcl-2 family of anti-apoptotic proteins, and the dasatinib–quercetin (DQ) cocktail, which inhibits tyrosine kinases and AKT signaling, among other pathways. We also assess the natural killer cell line NK92 to model emerging immune-mediated senescent cell ablation therapies. In synaptically mature cultures, we show that NK92 cells preferentially—but not exclusively—targeted A β -treated neurons and astrocytes with senescent-like phenotypes. DQ demonstrated a safe profile for human neurons, but Navitoclax exhibited non-selective neurotoxicity. These findings highlight potential risks associated with developing BBB-permeable therapies based on the mechanisms of NAV and NK cell-mediated cytotoxicity. Our work underscores the critical need for human-relevant models in the AD drug-development pipeline to improve safety and clinical translatability.

Keywords Alzheimer's disease, Cellular senescence, Senolytics, Primary human neurons, Primary human astrocytes, Neurodegeneration, Amyloid beta, Navitoclax, Dasatinib, Quercetin, NK92

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Introduction

The Food and Drug Administration (FDA) has approved three A β -clearing antibodies that, while effective at reducing biological markers of pathology, are yet to robustly stop clinicopathological AD progression [1–4]. Historically, a paradoxical disconnect has existed between positive results in preclinical research and the extraordinarily high failure rate in AD clinical trials [5–7]. This highlights two critical bottlenecks: the need for new therapies and the lack of translation from research in animal models to humans.

In this context, cellular senescence has emerged as a novel druggable target in AD [8]. Senescent cells, which accumulate with age, adopt a harmful senescence-associated secretory phenotype (SASP) that drives inflammation and contributes to tissue damage [9, 10]. Both tau and A β have been associated to deleterious cellular senescence in transgenic mouse models of AD neuropathology [11–13]. The selective elimination of senescent cells has demonstrated beneficial effects in preclinical studies, initially in mouse models of Parkinson's disease [14]. Subsequently senolytics, a class of drugs used to eliminate senescent cells [15] have shown promise in mitigating AD and AD-like pathology in rodent models [11–13, 16, 17] and have progressed to clinical trials for AD [8]. Importantly, different senolytics act through distinct mechanisms and can target different subsets of senescent cells [15]. In a transgenic mouse model of AD-like tau pathology, the senolytic cocktail Dasatinib plus Quercetin (DQ) reduced neurofibrillary tangle (NFT) burden, neuronal loss, and ventricular enlargement [11]. In an A β -based AD mouse model, DQ selectively eliminated senescent cells localized near A β plaques, decreased A β burden, suppressed inflammation, and improved cognitive performance [13]. In another tauopathy model, treatment with Navitoclax (NAV)—a Bcl-2 family inhibitor—prevented the upregulation of senescence-associated genes and reduced tau phosphorylation [12].

Senolytics are broadly categorized under a unifying umbrella and, as noted, demonstrate diverse mechanisms of action [18]. Perhaps unsurprisingly, the use of different senolytics in different models of AD-like pathology has resulted in conflicting evidence regarding which central nervous system (CNS) cell types—neurons, astrocytes, microglia, or oligodendrocyte progenitor cells (OPCs)—undergo deleterious cellular senescence which can be targeted by senolytics [11–13, 16–18]. Indeed, different senolytics have been reported to eliminate distinct, non-overlapping senescent non-CNS cell populations [18].

Laboratory-generated human neurons, such as those derived from induced pluripotent stem cells (iPSCs), are powerful tools commonly used to address species-specific differences inherent in the use of rodent models [19, 20]. However, the developmental origins and cellular

characteristics of laboratory-generated neurons differ fundamentally from those of bona fide human CNS neurons [21]. Specifically, iPSC-derived neurons originate from reprogrammed somatic cells and are directed through artificial differentiation protocols. In contrast, primary human neurons are derived directly from human CNS tissue and undergo their natural, lineage-specific developmental trajectory toward terminal differentiation. As such, primary neurons are considered an *in vitro* gold standard representing unaltered human CNS cells. As mentioned above, senolytics exhibit cell-type-specific toxicity [18]. Whether artificial reprogramming alters molecular pathways in laboratory-generated neurons in ways that affect senolytic selectivity and off-target toxicity remains unknown. To the best of our knowledge, such validation has not yet been performed.

Human primary neuron cultures have contributed to landmark insights into AD pathobiology [22–27] and have been extensively characterized for maturation and cell composition. Neurons begin to polarize and extend processes within the first week, with up to 90% of cells expressing class III β -tubulin by day 7 in cultures derived from 8–12-week tissue [28]. The microtubule associated protein 2 (MAP2) and synaptic markers such as synaptophysin increase by 28 days *in vitro* (DIV), accompanied by extensive neurite networks and electron microscopy evidence of mature synapses [29]. Synaptosomal-associated protein of 25 kDa (SNAP-25) and (postsynaptic density protein 95) (PSD-95) levels raise between 20 and 28 DIV, reflecting both presynaptic and postsynaptic maturation [30]. Kerkovich and colleagues demonstrate increasing outward delayed rectifying currents from 4 to 21 DIV, and the appearance of inward currents by 7 DIV [31]. Calcium imaging shows KCl-evoked Ca²⁺ influx by 4 DIV and sustained responses through at least 28 DIV [30]. The glutamatergic system also undergoes maturational changes: 1 mM glutamate has minimal toxicity up to 30 DIV but becomes neurotoxic by 35 DIV [32].

Cellular diversity in primary human CNS cultures evolves dynamically over time. In early post-replating stages, neurons constitute the majority cell type, comprising ~72.5% at 3 days, ~79.9% at 6 days, and ~73.7% at 20 days *in vitro*, as identified by class III β -tubulin and MAP2 immunostaining [31]. During the first two weeks *in vitro*, many cells simultaneously express Nestin, a neural stem/progenitor marker, glial fibrillary acidic protein (GFAP), an astrocytic marker, and pan-neuronal antigens labeling soma and neurites [30, 33]. This co-expression profile indicates the presence of bipotential or undifferentiated neural progenitors actively undergoing lineage specification. By approximately 24 DIV, this transitional state resolves into more defined cell identities, with segregation between GFAP-positive astrocytes and pan-neuronal marker-positive neurons, signaling commitment

to distinct glial and neuronal lineages [30]. Astrocytes, marked by GFAP, remain consistently low, ranging from ~5.3–6.8% across 3–20 DIV [31] but reports indicate pronounced increased GFAP staining from 24 DIV onwards [30]. Additional cell types such as parvalbumin⁺ interneurons, galactocerebroside-positive oligodendrocyte-lineage cells, and HLA-DR-positive microglia remain rare at 28 DIV, and markers of proliferation (Ki-67) and apoptosis (TUNEL) are consistently low across time points [29].

Human primary neuron cultures are a well characterized alternative which offer a biologically relevant model for studying age-related diseases including AD [21]. However, protocols that support neuronal maturation—an essential requirement for modeling the adult human brain—involve initiating cultures after tissue harvest, necessitating access to specialized medical facilities. This logistical barrier has limited the accessibility of mature human primary neuron cultures for modeling AD and related interventions in most laboratories. Cryopreservation offers a practical alternative by enabling the storage and shipment of human primary neurons. This approach was demonstrated to be feasible decades ago [34–36] and today, cryopreserved primary human neurons are commercially available. However, cryopreserved neurons typically exhibit reduced viability [35] which limits their capacity to mature and, by extension, their utility as models of the adult human brain. This limitation may explain why most AD models using mature human primary neurons, including models based on A β treatments as described here, are sparse and have remained confined to a few laboratories over several decades [22–[28, 33]37–[47].

By applying standard primary mouse neuron culture techniques and widely available modern reagents, we demonstrate that mixed primary human neuron and astrocyte cultures—derived from commercially available cryopreserved preparations—can be reliably maintained and matured in vitro. Using this system, we evaluated the two senolytic drug regimens previously preclinically explored as AD interventions—NAV and the senolytic cocktail DQ—in an in vitro model of A β -induced AD pathology [8]. NAV inhibits the Bcl-2 family of anti-apoptotic proteins [18, 48–50]. While Dasatinib inhibits tyrosine kinases, including Src family kinases and Bcr-Abl [51, 52] Quercetin targets survival pathways including Bcl-2 and AKT signaling [53, 54]. To benchmark NAV and DQ, we compare these senolytics to a model of physiological senescent cell ablation using the natural killer (NK) cell line NK92, modeling immune system-mediated clearance [55, 56]. The results for NK92 co-culture experiments are consistent with the preferential ablation of senescent-like populations amidst non-selective elimination of healthy-control cells. DQ did not show neurotoxicity and only one of several metrics showed a

non-selective impact on astrocyte viability. In contrast, NAV exhibited dose-dependent toxicity in both healthy and A β -treated neurons. These findings underscore the need to refine senolytic strategies in human-relevant models of AD.

Results

Characterization of cells types and maturation of cryoconserved human neuronal-astrocytic cultures

Only minor modifications to the vendor-provided culture protocol and commonly used mouse primary neuron culture techniques were required to successfully maintain these cells for over 100 DIV. Briefly, the vendor recommends a media wash within the first 24 h to remove dimethyl sulfoxide (DMSO) from the cryopreservation solution; however, we found this step to be detrimental to long-term cell viability (data not shown). Instead, we allowed the cells to remain undisturbed in the culture medium for 48 h, after which the DMSO was further and gradually diluted through scheduled media refreshment. Critically, we avoided complete media changes, instead replacing approximately one-third of the medium every two days to retain a portion of the conditioned medium, which contains cell-derived trophic factors. It is important not to remove a fixed volume (e.g., 100 μ L from a 300 μ L well), as media evaporation over time reduces the total volume, and fixed-volume aspiration may eventually expose cells to air and cause desiccation. Instead, it is essential that the media be aspirated only to a visually determined two-thirds minimum volume, followed by the addition of a one-third fresh medium. Evaporation must also be carefully monitored, as this approach inherently leads to a gradual irreversible increase in solute concentration over time. We recommend seeding cells at a density of $1\text{--}1.5 \times 10^4$ cells/cm², a range in which cytosine arabinoside (Ara-C) is not needed to curb astrocyte proliferation. This was of particular importance for our studies, as Ara-C is a DNA-damaging agent that could induce cellular senescence. However, we did test Ara-C and it was tolerated (data not shown). While we did not attempt this approach, we think it is possible to maintain very low-density primary neuron cultures supported by an astrocytic feeder layer, as previously described [57]. By following the straightforward steps outlined in the Methods section, users should be able to maintain human primary neuron and astrocyte cultures beyond 100 DIV.

The cells from primary human cell preparations isolated from commercially available human embryonic cells extend processes (Videos 1, 2), undergo nuclear migration (Video 3), and cell division (Videos 3, 4) within 48 h after plating. Based on morphology, different cell types are identifiable shortly after plating (Video 5). Neurons positive for the neuronal markers NeuN and MAP2 as well as GFAP-positive astrocytes are present early on

(Fig. 1a). Previous studies have reported the separation of pan-neuronal markers from Nestin and GFAP, indicating lineage commitment of neuronal cells [30]. Consistent with this, we observed that GFAP-positive astrocytes also express S100 β —a marker of mature astrocytes—at both 28 and 43 DIV (Sup. Figure 1a, b, c). In addition, we noted the presence of smaller S100 β -positive astrocytes with less elaborate processes that lacked detectable GFAP expression, a population previously described in the literature [58]. These cultures can be maintained for 100 DIV (Fig. 1a, b) and we have successfully cultured up to 200 DIV (Sup. Figure 1 d); the upper age limit for these cultures is currently unknown. Human primary neurons present characteristic markers including a dense mesh of tau 3R-positive axons (Fig. 1c), positive immunostaining for the pan-tau marker tau-5 (Sup. Figure 1 e) and for Ankyrin-G (Sup. Figure 1d, f), a marker for axon initial segments [59].

We observed several unexpected phenotypes in these cultures, with arguably the most striking being their extended viability (Sup. Figure 1d). Another unexpected finding was that NeuN staining was not restricted to perinuclear regions but also extended into processes during maturation (Fig. 1a, b). We also detected soma staining with MN423 (Fig. 1c), a tau antibody purportedly exclusively recognizing the pronase-resistant core of tau paired helical filaments (PHF), but not full-length human tau [60, 61]. Additionally, some axons were positive for MC-1 antibody (Sup. Figure 1g), which was previously thought to exclusively detect pathological tau conformations associated with neurofibrillary tangles [62].

The expression of several neuronal maturation markers and the establishment of a mature cell-type composition typically plateau after approximately three weeks in culture [30, 32, 33, 39]. Although the exact timing can vary depending on factors such as neuronal plating density, we selected this timeframe to represent a consistent stage of neuronal maturity for initiating our experiments, marked by the exogenous administration of A β . Regarding the experimental duration, the development of senescent phenotypes has been reported to occur over approximately 5–14 days, depending on the cell type and the method used to induce cellular senescence [9, 63, 64]. Accordingly, we allowed a 13-day period following A β treatment before applying senolytic regimens for 24 h.

Prior to embarking on these analyses, we first sought to validate that the maturation trajectory of our cryopreserved cultures aligned with those previously reported in non-cryopreserved cultures [30, 32, 33, 39]. To this end, we assessed synaptic markers at 14 DIV as a benchmark for immature cultures, at 28 DIV to confirm the maturity stage at which A β treatment was planned be applied, and at 43 DIV to evaluate the maturity status at the conclusion of the senolytic regimens at 42 DIV. We performed

immunocytochemistry (ICC) for the presynaptic marker synapsin I (synapsin) and postsynaptic marker Homer1a-d (Homer1; Fig. 2a), as well as synapsin and the postsynaptic marker PSD95 (Fig. 2b) at 14, 28, and 43 DIV. Synapsin is a vesicle-associated protein localized at axon terminals, while Homer1 and PSD95 are scaffolding proteins found at the postsynaptic density [65, 66].

A one-way ANOVA revealed significant changes in synapsin expression with maturation ($p < 0.001$; Fig. 2c; see Table 1a for omnibus test results). Multiple comparisons showed that synapsin intensity at 14 DIV was significantly lower than at 28 ($p < 0.001$) and 43 DIV ($p < 0.001$; Fig. 2c). However, the difference between 28 DIV and 43 DIV was not statistically significant ($p < 0.163$; Fig. 2c). Likewise, significant differences were also observed for Homer1 expression across DIV ($p = 0.004$; Table 1b), with significant increases in intensity between 14 and 28 DIV ($p = 0.010$) and between 14 and 43 DIV ($p = 0.004$), but not between 28 and 43 DIV ($p = 0.667$; Fig. 2d). These findings indicate a significant increase in both pre and postsynaptic markers between 14 and 28 DIV and between 14 and 43 DIV, reflecting robust synaptic development during early neuronal maturation, with a plateau in development observed between 28 and 43 DIV.

As per the analysis of PSD95 detection and time point of maturation, a one-way ANOVA showed highly significant differences between DIV of culture ($p < 0.001$; Table 1c; Fig. 2e). Multiple comparisons aimed at identifying specific DIV with significant differences revealed only a non-significant trend toward an increase in PSD95 expression between DIV 14 and DIV 28 ($p = 0.062$; Fig. 2e). However, the differences were significant when comparing 14 and 43 DIV ($p < 0.001$) as well as 28 and 43 DIV ($p = 0.005$; Fig. 2e). These findings indicate ongoing postsynaptic development between 14 and 43 DIV.

To further characterize synaptic maturity, we assessed the colocalization of Homer1 with Synapsin and PSD95 with Synapsin for each time point. Colocalized Homer1 and Synapsin puncta showed significant differences ($p < 0.001$; Table 1d), with a non-significant trend towards increased colocalization from 14 to 28 DIV ($p = 0.093$) and a significant increase between 14 and 43 DIV ($p < 0.001$; Fig. 2f). The differences between 28 and 43 DIV were not significant ($p = 0.771$; Fig. 2f). As per colocalized PSD95 and Synapsin puncta, there were also highly statistically significant differences with maturation ($p < 0.001$; Table 1e; Fig. 2g). There was a significant increase from 14 to 28 DIV ($p = 0.001$) and from 14 to 43 ($p < 0.001$) DIV but not from 28 to 43 DIV ($p = 0.102$; Fig. 2g). These results indicate significant increases in Homer1-Synapsin and PSD95-Synapsin colocalization during early neuronal maturation, with a plateau observed between 28 and 43 DIV. Collectively, these results support the conclusion that the selected

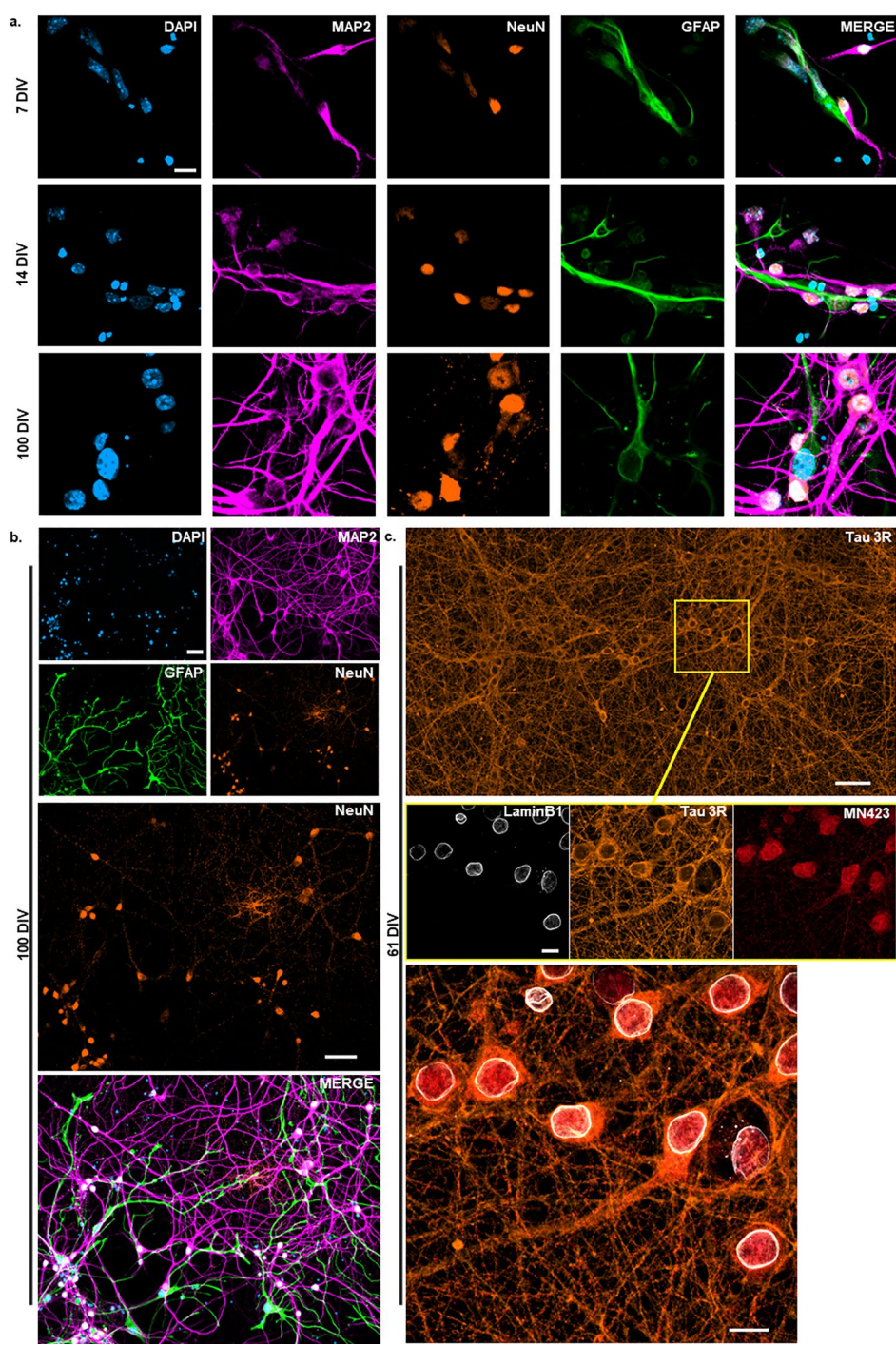


Fig. 1 Development of neuronal and astrocytic markers in human primary neurons during maturation. **(a)** Super-resolution confocal images of cultures at 7, 14, and 100 DIV. DAPI (nuclei), MAP2 (dendrites), NeuN (neuron somas), and GFAP (astrocytes). The increasing complexity of MAP2 dendritic arbors reflects neuronal maturation. At 100 DIV, NeuN puncta extends beyond somatodendritic regions. 63x objective with 1.7x zoom. Scale bar = 10 μ m. **(b)** Confocal tile scan of 100 DIV cultures with immunostaining as in **(a)**. 20x objective with 1x zoom. Scale bar = 50 μ m. **(c)** Confocal tile scan (top) of 61 DIV cultures with tau stained by tau 3R antibody. The detailed images (middle) focus on a region of the tile scan (yellow box) with immunostaining for LaminB1 (nuclear lamina), Tau 3R (axons), and MN423 (tau PHF). The enlarged image (bottom) merges LaminB1, tau 3R, and MN423. 40x objective with 2x zoom. Scale bars: tile scan (top) = 50 μ m; detail images (middle) and merged images (bottom) = 10 μ m

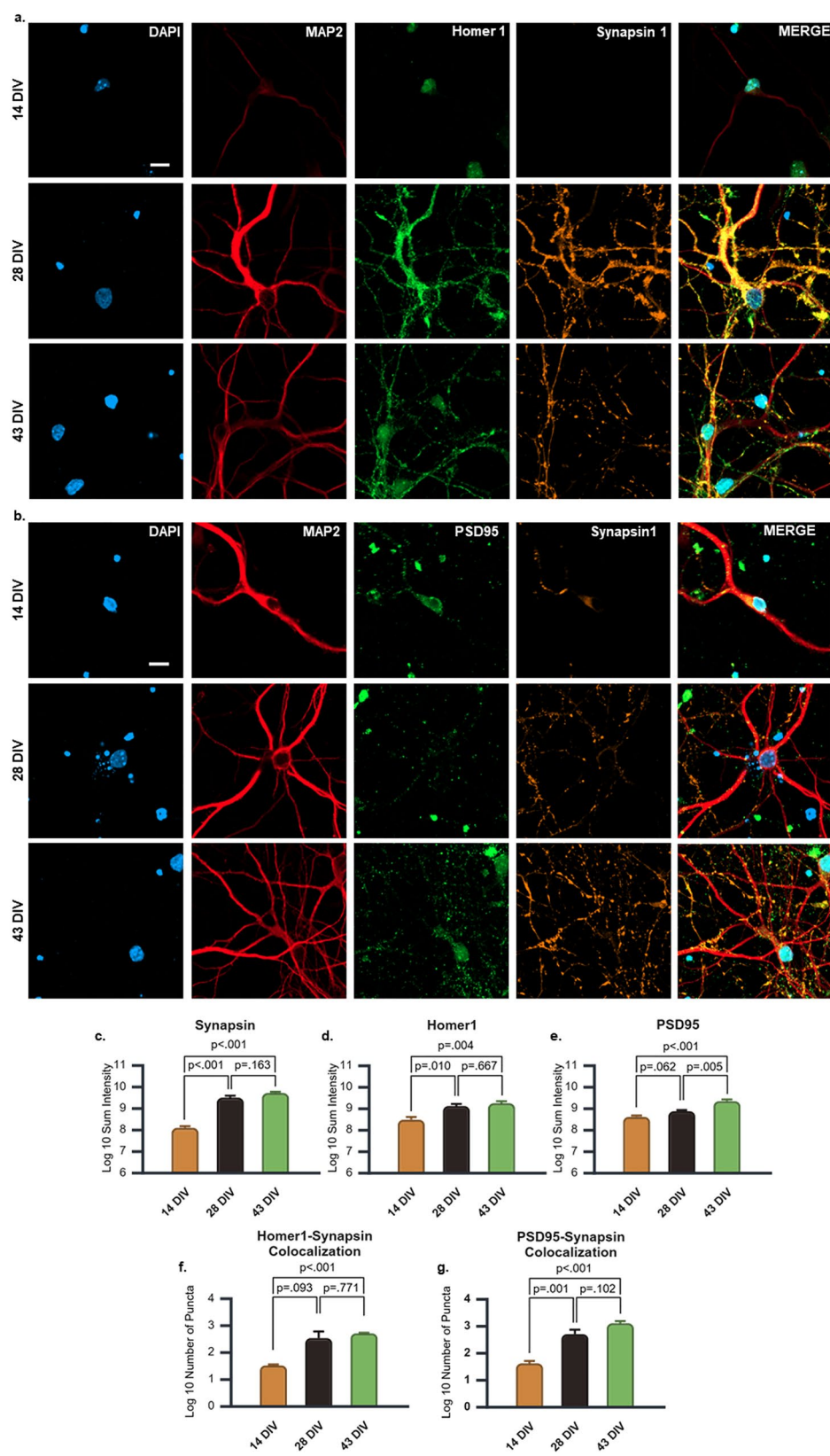


Fig. 2 (See legend on next page.)

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Fig. 2 Synaptic markers in human primary neurons during maturation. **(a, b)** Representative super-resolution confocal images from cultures at 14, 28, and 43 DIV with staining for DAPI, MAP2 with **(a)** Homer1 (postsynaptic marker) plus Synapsin (presynaptic marker) or **(b)** PSD95 (postsynaptic marker) plus Synapsin. DAPI was used to identify nuclei, MAP2 to label somatodendritic neuronal processes. 63x objective with 2x zoom. Scale bar = 10 μ m. **(c)** Quantification of log10-transformed total Synapsin puncta signal intensity at 14, 28, and 43 DIV. Three biological replicates, each with 2 technical replicates. **(d, e)** Quantification of log10-transformed total **(d)** Homer1 and **(e)** PSD95 puncta signal intensity at 14, 28, and 43 DIV. **(f, g)** Quantification of log10-transformed number of colocalized puncta of **(f)** Homer1-Synapsin and **(g)** PSD95-Synapsin at 14, 28, and 43 DIV. Three biological replicates. **(c–e, g)** Tukey and **(f)** Games-Howell all pairwise comparisons. Significance, $p < 0.05$. Graphs display mean $\pm$ s.e.m

intervention time points were appropriate for evaluating senolytic regimens involving NK92 cells, NAV, and DQ.

Numerous previous studies have modeled AD in vitro following exogenous administration of A β [22, 26, 27, 37, 39, 40, 42, 46, 47]. These experiments are typically conducted in two-dimensional (2D) cultures rather than three-dimensional (3D) systems in which cells are embedded in hydrogel scaffolds (e.g., PuraMatrix or Matrigel). In 2D cultures, monomeric A β secreted by cells engineered to overexpressing familial AD (FAD)-linked genes can diffuse into the media and may be lost during routine media changes [67]. For these in vitro FAD models, 3D scaffolds may enhance A β retention and, consequently, promote aggregation. Conversely, 3D scaffolds may hinder the accessibility of exogenously administered, pre-aggregated A β species, particularly larger oligomers and fibrils. Therefore, we chose to employ traditional 2D cultures for our experiments.

To determine whether exogenous A β persists in 2D cultures following extended media replacement, we treated human primary neurons at 28 DIV with A β (1–42) and assessed whether aggregates were still present at 61 DIV. A β (1–42) preparations enriched in oligomers (0.5 μ M) could be readily detected 33 days post-treatment (DPT), as demonstrated by immunostaining with the 4G8 antibody (Sup. Figure 2a), which targets residues 17–24 of A β . This was confirmed by the conformation-specific OC antibody (Sup. Figure 2 b), which recognizes fibrillar A β oligomers and fibrils [68]. Preparations enriched in fibrils (0.5 μ M) were similarly positive 33 DPT (Sup. Figure 2b, bottom panel).

To confirm marker specificity, we assessed the widely used A β /APP-targeting antibody 6E10 together with 4G8 and OC in cell-free conditions (Sup. Figure 2c). DAPI was also included, based on observations that DAPI stained cell nuclei but also some A β (1–42) aggregates in primary neuron cultures. As expected, all antibodies co-localized on A β (1–42) aggregates (Sup. Figure 2c). Notably, some larger plaques exhibited a shell-like pattern with peripheral antibody staining and a DAPI-positive core. This may indicate that antibody penetration is limited in denser aggregates, whereas DAPI remains able to access the plaque interior.

Having confirmed the persistence of A β aggregates following a single administration and characterizing the maturation profile of the neurons in culture, we implemented a protocol in which A β was applied at 28 DIV,

followed by a 13-day incubation period prior to the introduction of NK92 cells, NAV, or DQ. The effects of these interventions were then assessed at 43 DIV. These treatments were chosen to target overlapping and distinct senescence-associated pathways, with the aim of gaining mechanistic insight that could inform the rational design of future, more selective senolytic strategies (Sup. Figure 2d).

NK92 co-culture results in cell killing compatible with the Preferential elimination of senescent-like neurons and astrocytes in an A β in vitro model of AD

The immune system naturally eliminates senescent cells through senescence immune surveillance, primarily mediated by NK cells, CD8+ cytotoxic T-cells (CTT), and macrophages [55]. In vitro studies have utilized NK92 cells, a human NK cell line, to model the immune-mediated clearance of senescent cells [56]. To investigate the potential of senescence immune surveillance as a therapeutic for AD as well as to benchmark NAV and DQ, we co-cultured NK92 cells with primary neurons and astrocytes at 42 DIV for 24 h and assessed whether there were differences in selective killing between A β (1–42)-treated (0.5 μ M, 28 DIV) and vehicle-treated (NaOH) cells (Fig. 3a). A β (1–42) preparations were enriched in oligomers, and a low concentration of 0.5 μ M was chosen from ranges reported in the literature to allow evaluation of senolytic activity at 43 DIV while minimizing short-term toxicity unrelated to cellular senescence [27, 37].

First, we performed exploratory time-lapse imaging experiments to track live cell behavior in the absence of NK92 cells (Video 6) and in co-culture with NK92 cells for both non-A β (Video 7) and A β -treated cultures (Video 8; Sup. Figure 3a). NK92 cells were viable in neurobasal medium, formed uropods, and actively interacted with target neurons and astrocytes irrespective of A β treatments (Videos 7 and 8). Having established the feasibility of NK92 co-cultures, we used confocal spectral scanning with linear unmixing to determine whether the effects of NK92 co-culture were consistent with selective senescent cell ablation. This imaging technique allowed us to simultaneously resolve the 6 channels resulting from immunocytochemistry (ICC) for DAPI to identify nuclei, HMGB1 and LaminB1 as markers of cellular senescence [10, 69–71] active caspase-3 (aCasp3) to identify apoptotic cells [72] as well as MAP2 and GFAP to distinguish neurons and astrocytes, respectively (Fig. 3b;

Table 1 Synapse maturation

	Variable	Effects	Min Shapiro-Wilk	Levene's test	Test results for ANOVA and Mean Differences	
a	Synapsin	DIV	0.326	0.645	ANOVA	$F(2, 6) = 152.466, p < 0.001, \eta^2 = 0.981$
		28 Vs. 14 DIV			Tukey-HSD	1.4033; 95% CI, 1.0945 to 1.7121, $p < 0.001$
		43 Vs. 14 DIV			Tukey-HSD	1.6180; 95% CI, 1.3092 to 1.9268, $p < 0.001$
		43 Vs. 28 DIV			Tukey-HSD	0.2147; 95% CI, -0.0941 to 0.5235, $p = 0.163$
b	Homer1	DIV (Homer)	0.052	0.601	ANOVA	$F(2, 6) = 16.410, p = 0.004, \eta^2 = 0.845$
		28 Vs. 14 DIV			Tukey-HSD	0.6380; 95% CI, 0.1988 to 1.0772, $p = 0.010$
		43 Vs. 14 DIV			Tukey-HSD	0.7651; 95% CI, 0.3260 to 1.2043, $p = 0.004$
		43 Vs. 28 DIV			Tukey-HSD	0.1271; 95% CI, -0.3120 to 0.5663, $p = 0.667$
c	PSD95	DIV	0.102	0.529	ANOVA	$F(2, 6) = 33.111, p < 0.001, \eta^2 = 0.917$
		28 Vs. 14 DIV			Tukey-HSD	0.2615; 95% CI, -0.0151 to 0.5381, $p = 0.062$
		43 Vs. 14 DIV			Tukey-HSD	0.7243; 95% CI, 0.4477 to 1.001, $p < 0.001$
		43 Vs. 28 DIV			Tukey-HSD	0.4628; 95% CI, 0.1862 to 0.7394, $p = 0.005$
d	Homer1	DIV	0.234	0.009	Welch F	$F(2, 3.161) = 473.591, p < 0.001$
	Synapsin I	28 Vs. 14 DIV			Games-Howell	1.0130; 95% CI, -0.3916 to 2.4177, $p = 0.093$
	Coloc.	43 Vs. 14 DIV			Games-Howell	1.1931; 95% CI, 1.0423 to 1.3440, $p < 0.001$
		43 Vs. 28 DIV			Games-Howell	0.1801; 95% CI, -1.2529 to 1.6131, $p = 0.771$
e	PSD95	DIV	0.152	0.192	ANOVA	$F(2, 6) = 45.180, p < 0.001, \eta^2 = 0.938$
	Synapsin I	28 Vs. 14 DIV			Tukey-HSD	1.0830; 95% CI, 0.5865 to 1.5796, $p = 0.001$
	Coloc.	43 Vs. 14 DIV			Tukey-HSD	1.4876; 95% CI, 0.9910 to 1.9841, $p < 0.001$
		43 Vs. 28 DIV			Tukey-HSD	0.4045; 95% CI, -0.0920 to 0.9011, $p = 0.102$

Sup. Figure 3b, c; Video 9). Confocal tile scans covering several hundred square microns were acquired across three replicates for each treatment condition (Sup. Figure 3d; Video 9).

For the statistical analyses of marker expression, two-way ANOVAs with interaction terms were conducted when the appropriate parametric assumptions were met. A detailed explanation of the statistical workflow, along with a basic overview of two-way ANOVAs with interaction terms, is provided in the Materials and Methods section under Statistics. The interaction term, main effects, and simple main effects of the two-way ANOVA as well as the corrections for familywise error are conceptually explained the first time they appear in the text. To enhance clarity, interaction effects and main effects are reported and graphed only when statistically significant, while simple main effects are always displayed. However, the complete statistical results are available in the tables referenced in each figure.

We first examined whether NK92 co-culture affected neuronal viability by analyzing the number of cells categorized as live neurons in each condition. A two-way ANOVA showed there was a significant main effect for A β treatment, revealing a reduction in the number of viable neurons with A β (1–42) treatment ($p = 0.013$, Fig. 3c, d; Table 2a). After establishing a Bonferroni corrected threshold for two simple main effects ($p < 0.025$), the analysis of the simple main effects did not show significant differences between treatment conditions (Fig. 3d; Table 2a).

In neurons, we separately assessed cytoplasmic and nuclear aCasp3 to distinguish initial activation from subsequent nuclear translocation during apoptosis [73, 74]. Confocal images of aCasp3 showed that NK92 cells likely underwent apoptosis (Fig. 3b; Sup. Figure 3b, c), therefore the analysis was restricted to live neurons and astrocytes positive for MAP2 and GFAP, respectively. For neurons, the main effects for NK92 were statistically significant, revealing increased nuclear aCasp3 with NK92 co-culture irrespective of A β treatment ($p = 0.046$, Fig. 3e; Table 2b). Bonferroni corrected simple main effects ($p < 0.025$) did not show significant differences between treatment conditions (Fig. 3f; Table 2b).

The analysis of aCasp3 within MAP2-positive neuronal processes also increased significantly in the presence of NK92 co-culture irrespective of A β treatment ($p = 0.045$, Fig. 3g; Table 2c), revealing non-selective effects. No significant differences or trends were found between treatment groups (Fig. 3h), indicating cytoplasmic aCasp3 was not preferentially accumulating in A β -treated cells. Notably, in confocal images a clear aCasp3 signal was not visually discernible in MAP2-positive processes, suggesting that the apoptotic signal detected in the analysis reflects subtle changes. Overall, the analyses of nuclear ($p = 0.046$) and somatodendritic aCasp3 ($p = 0.045$) indicate a non-selective increase of cell ablation with NK92 co-culture.

It is well established that there is no universal marker of cellular senescence [75]. Numerous senescence-associated markers have been examined in the CNS, including

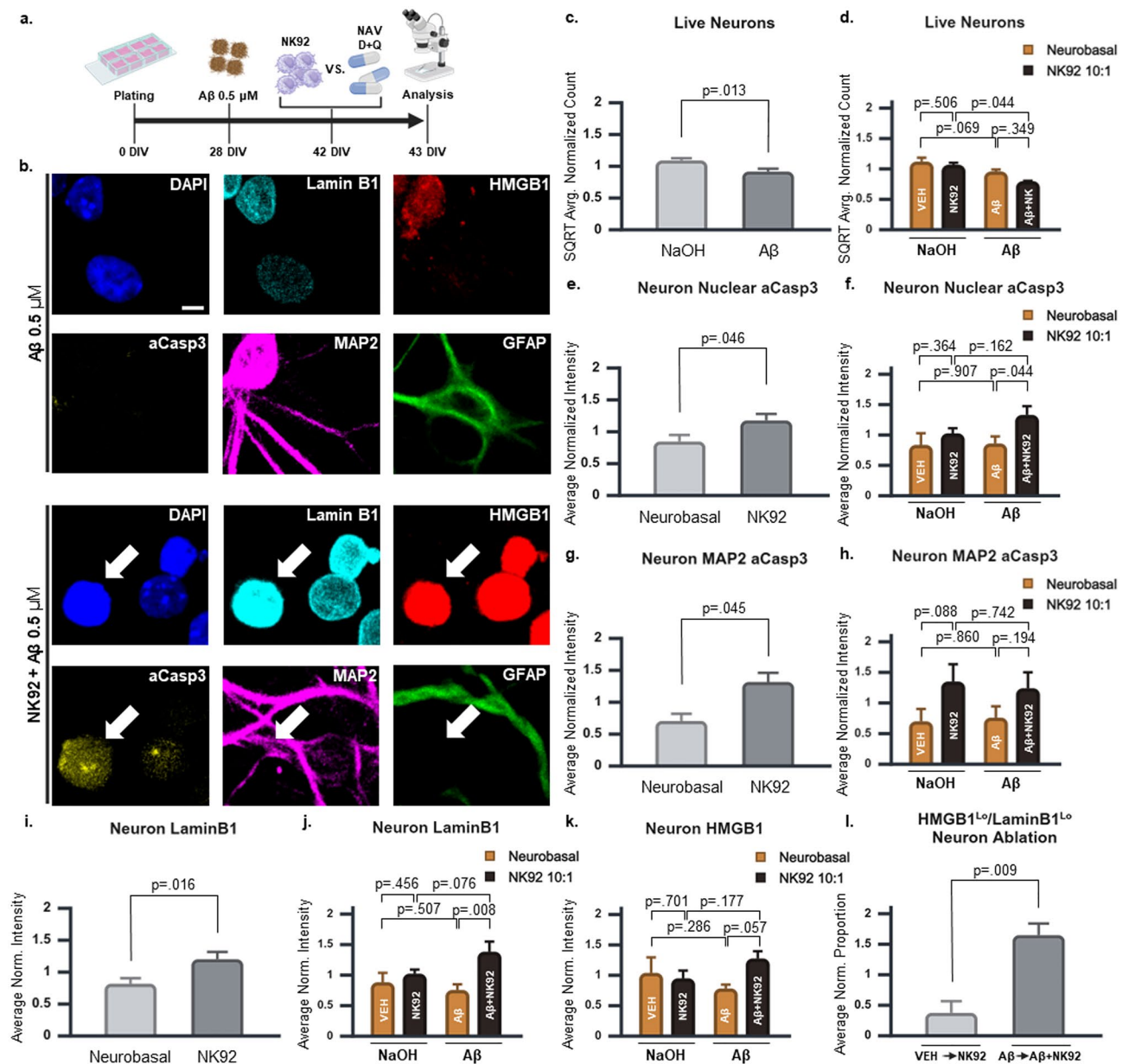


Fig. 3 In vitro AD model of senescence immune surveillance in human neurons. **(a)** Diagram depicting human in vitro model of AD to assess senescent cell ablation by NK92, NAV, and DQ. Primary mixed human neuron and astrocyte cultures were treated with either vehicle (NaOH) or 0.5 μM of Aβ preparations enriched in oligomers at 28 DIV. At 42 DIV, NK92 cells, NAV or DQ were added to the cultures. Cells were fixed for ICC at 43 DIV. **(b)** Representative linearly unmixed confocal spectral scans of Aβ (top 2 rows) and Aβ + NK92 (bottom two rows) groups (also see Sup. Figure 3b). The white arrow identifies a putative NK92 cell that is MAP2⁺/GFAP⁺/aCasp3⁺. 40x objective with 1x zoom. Scale bar = 5 μm. **(c)** Aβ main effects and **(d)** Aβ x NK92 simple main effects on square root (SQRT) transformed average normalized viable neuron count. The data was SQRT-transformed to meet the assumption of normality (Table 2a). **(e)** NK92 main effect and **(f)** Aβ x NK92 simple main effects on nuclear aCasp3. **(g)** NK92 main effect and **(h)** Aβ x NK92 simple main effects on aCasp3 within MAP2 processes. **(i)** NK92 main effect and **(j)** Aβ x NK92 simple main effects on LaminB1. **(k)** Aβ x NK92 simple main effects on HMGB1. **(l)** Comparison of HMGB1^{Lo}/LaminB1^{Lo} neuronal loss between VEH→NK92 and Aβ→Aβ + NK92, quantified as described in Sup. Figure 3e. **(c, e, g, i)** Main effects. Significance, $p < 0.05$. **(d, f, h, j, k)** Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. **(l)** Independent sample *t*-test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. 3x Biological Replicates. A total of 6111 neurons were analyzed. Graphs represent mean ± s.e.m

in contexts that do not directly aim to assess senescence [76]. In CNS cells, interpretation is further complicated by the fact that most neurons are postmitotic, precluding the use of proliferation-dependent markers such as Ki67

and others. Moreover, one of the most widely used senescence markers, senescence-associated β-galactosidase (SA-β-gal) [77] has had its reliability in neurons repeatedly questioned [11, 78, 79]. Another classical marker

Table 2 NK92 co-culture (10:1 effector to target Ratio)

Variable	Interaction, main and simple effects		Min Shapiro-Wilk	Levene's test	ANOVA/Kruskal-Wallis/t-test
a Neuron Viability:	Interaction		(0.048*) 0.056	0.099	$F(1, 8) = 0.045, p = 0.838$, partial $\eta^2 = 0.006$
Square root	A β O				$F(1, 8) = 10.072, p = 0.013$, partial $\eta^2 = 0.557$
transform of	NK92				$F(1, 8) = 1.427, p = 0.266$, partial $\eta^2 = 0.151$
average normalized	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 4.388, p = 0.069$, partial $\eta^2 = 0.354$
total cell count	NK92	NaOH Vs. A β O			$F(1, 8) = 5.728, p = 0.044$, partial $\eta^2 = 0.417$
to amend violation	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.484, p = 0.506$, partial $\eta^2 = 0.057$
of normality*	A β O	Neurobasal Vs. NK92			$F(1, 8) = 0.988, p = 0.349$, partial $\eta^2 = 0.110$
b Neuron Nuclear	Interaction		0.175	0.551	$F(1, 8) = 1.009, p = 0.345$, partial $\eta^2 = 0.112$
aCasp3:	A β O				$F(1, 8) = 1.381, p = 0.274$, partial $\eta^2 = 0.147$
Average normalized	NK92				$F(1, 8) = 5.959, p = 0.046$, partial $\eta^2 = 0.412$
mean nuclear	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.015, p = 0.907$, partial $\eta^2 = 0.002$
aCasp3 signal	NK92	NaOH Vs. A β O			$F(1, 8) = 2.376, p = 0.162$, partial $\eta^2 = 0.229$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.926, p = 0.364$, partial $\eta^2 = 0.104$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 5.678, p = 0.044$, partial $\eta^2 = 0.415$
c aCasp3 within MAP2:	Interaction		0.133	0.78	$F(1, 8) = 0.136, p = 0.721$, partial $\eta^2 = 0.017$
Average normalized	A β O				$F(1, 8) = 0.013, p = 0.914$, partial $\eta^2 = 0.002$
aCasp3 signal per unit	NK92				$F(1, 8) = 5.647, p = 0.045$, partial $\eta^2 = 0.414$
volume of MAP2	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.033, p = 0.860$, partial $\eta^2 = 0.004$
	NK92	NaOH Vs. A β O			$F(1, 8) = 0.116, p = 0.742$, partial $\eta^2 = 0.014$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 3.769, p = 0.088$, partial $\eta^2 = 0.320$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 2.014, p = 0.194$, partial $\eta^2 = 0.201$
d Neuron LaminB1	Interaction		0.106	0.303	$F(1, 8) = 3.720, p = 0.090$, partial $\eta^2 = 0.317$
Average normalized	A β O				$F(1, 8) = 0.896, p = 0.371$, partial $\eta^2 = 0.101$
mean LaminB1 signal	NK92				$F(1, 8) = 9.223, p = 0.016$, partial $\eta^2 = 0.535$
	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.482, p = 0.507$, partial $\eta^2 = 0.057$
	NK92	NaOH Vs. A β O			$F(1, 8) = 4.134, p = 0.076$, partial $\eta^2 = 0.341$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.614, p = 0.456$, partial $\eta^2 = 0.071$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 12.329, p = 0.008$, partial $\eta^2 = 0.606$
e Neuron nuclear	Interaction		0.446	0.326	$F(1, 8) = 3.445, p = 0.101$, partial $\eta^2 = 0.301$
HMGB1:	A β O				$F(1, 8) = 0.058, p = 0.817$, partial $\eta^2 = 0.007$
Average normalized	NK92				$F(1, 8) = 1.672, p = 0.232$, partial $\eta^2 = 0.173$
mean nuclear	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 1.306, p = 0.286$, partial $\eta^2 = 0.140$
HMGB1 signal	NK92	NaOH Vs. A β O			$F(1, 8) = 2.196, p = 0.177$, partial $\eta^2 = 0.215$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.159, p = 0.701$, partial $\eta^2 = 0.019$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 4.958, p = 0.057$, partial $\eta^2 = 0.383$
f Neuron	NK92/A β O+NK92	0.276	1	$t(4) = 4.786, p = 0.009$	
HMGB1Lo/LaminB1Lo					
See Sup. Fig. 3e					
g Astrocyte Viability:	Interaction		0.051	0.366	$F(1, 8) = 0.012, p = 0.915$, partial $\eta^2 = 0.001$
Average normalized	A β O				$F(1, 8) = 3.514, p = 0.098$, partial $\eta^2 = 0.305$
total cell count	NK92				$F(1, 8) = 2.665, p = 0.141$, partial $\eta^2 = 0.250$
	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 1.968, p = 0.198$, partial $\eta^2 = 0.197$
	NK92	NaOH Vs. A β O			$F(1, 8) = 1.558, p = 0.247$, partial $\eta^2 = 0.163$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 1.517, p = 0.253$, partial $\eta^2 = 0.159$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 1.160, p = 0.313$, partial $\eta^2 = 0.127$
h Astrocyte Nuclear	Interaction		0.266	0.168	$F(1, 8) = 1.333, p = 0.282$, partial $\eta^2 = 0.143$
aCasp3:	A β O				$F(1, 8) = 0.645, p = 0.445$, partial $\eta^2 = 0.075$
Average normalized	NK92				$F(1, 8) = 11.584, p = 0.009$, partial $\eta^2 = 0.592$
mean nuclear	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.062, p = 0.810$, partial $\eta^2 = 0.008$
aCasp3 signal	NK92	NaOH Vs. A β O			$F(1, 8) = 1.917, p = 0.204$, partial $\eta^2 = 0.193$

Table 2 (continued)

Variable	Interaction, main and simple effects		Min Shapiro-Wilk	Levene's test	ANOVA/Kruskal-Wallis/t-test
i aCasp3 within GFAP: Average normalized aCasp3 signal per unit volume of GFAP	NaOH	Neurobasal Vs. NK92	0.376	0.160	$F(1, 8) = 2.529, p = 0.150$, partial $\eta^2 = 0.240$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 10.388, p = 0.012$, partial $\eta^2 = 0.565$
	Interaction				$F(1, 8) = 0.010, p = 0.922$, partial $\eta^2 = 0.001$
	A β O				$F(1, 8) = 0.512, p = 0.494$, partial $\eta^2 = 0.060$
	NK92				$F(1, 8) = 4.995, p = 0.056$, partial $\eta^2 = 0.384$
	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.189, p = 0.676$, partial $\eta^2 = 0.023$
	NK92	NaOH Vs. A β O			$F(1, 8) = 0.334, p = 0.579$, partial $\eta^2 = 0.040$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 2.275, p = 0.170$, partial $\eta^2 = 0.221$
j Astrocyte LaminB1 Average normalized mean LaminB1 signal	A β O	Neurobasal Vs. NK92	0.490	0.459	$F(1, 8) = 2.729, p = 0.137$, partial $\eta^2 = 0.254$
	Interaction				$F(1, 8) = 2.058, p = 0.189$, partial $\eta^2 = 0.205$
	A β O				$F(1, 8) = 0.153, p = 0.706$, partial $\eta^2 = 0.019$
	NK92				$F(1, 8) = 4.326, p = 0.071$, partial $\eta^2 = 0.351$
	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 0.544, p = 0.482$, partial $\eta^2 = 0.064$
	NK92	NaOH Vs. A β O			$F(1, 8) = 1.667, p = 0.233$, partial $\eta^2 = 0.172$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.208, p = 0.660$, partial $\eta^2 = 0.025$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 6.176, p = 0.038$, partial $\eta^2 = 0.436$
k Astrocyte nuclear HMGB1: Average normalized mean nuclear HMGB1 signal	Interaction		0.327	0.080	$F(1, 8) = 2.413, p = 0.159$, partial $\eta^2 = 0.232$
	A β O				$F(1, 8) = 0.507, p = 0.497$, partial $\eta^2 = 0.060$
	NK92				$F(1, 8) = 0.395, p = 0.547$, partial $\eta^2 = 0.047$
	Neurobasal	NaOH Vs. A β O			$F(1, 8) = 2.556, p = 0.148$, partial $\eta^2 = 0.243$
	NK92	NaOH Vs. A β O			$F(1, 8) = 0.354, p = 0.568$, partial $\eta^2 = 0.042$
	NaOH	Neurobasal Vs. NK92			$F(1, 8) = 0.428, p = 0.531$, partial $\eta^2 = 0.051$
	A β O	Neurobasal Vs. NK92			$F(1, 8) = 2.380, p = 0.161$, partial $\eta^2 = 0.229$
	NK92/A β O+NK92				$t(4) = 12.169, p < 0.001$
l Astrocyte HMGB1Lo/LaminB1Lo See Sup. Fig.3e			0.087	1	

is p16^{Ink4a}, a cyclin-dependent kinase inhibitor directly involved in the negative regulation of the cell cycle [75]. However, the antibodies we tested at the time of performing the experiments were unreliable (data not shown), consistent with the existing literature [75].

The decrease in the levels of Lamin B1 and nuclear HMGB1 have emerged as alternate reliable senescence-associated markers in the CNS. These markers have been used by ourselves and others to assess cellular senescence in vivo in human and mouse brain tissue [14, 80–82]. In senescent cells, levels of both Lamin B1 and/or nuclear HMGB1 have been shown to be reduced [10]69–[71, 75, 83, 84]. Lamin B1 is a structural component of the nuclear lamina and essential for brain development [85]. However, its decrease during senescence destabilizes nuclear architecture, contributing to chromatin remodeling, activation of inflammatory signaling, and the maintenance of the senescent state [70, 71, 83, 84]. HMGB1, a highly conserved, ubiquitously expressed nuclear protein [86]. In senescent cells, HMGB1 is actively translocated to the cytoplasm and secreted into the extracellular environment, where it acts as a damage-associated molecular pattern or DAMP [69]. Taken together, the reduction of LaminB1 and nuclear HMGB1 can be used to identify

senescent cells. Conversely, the selective elimination of a senescent subpopulation—defined by reduced Lamin B1 and/or nuclear HMGB1—would be expected to increase the overall population mean for these markers, as the remaining non-senescent cells would drive a shift toward their higher LaminB1 and nuclear HMGB1 populations means.

With the above in mind, we proceeded with the analyses of LaminB1 and nuclear HMGB1 in neurons and astrocytes. For neurons, NK92 co-culture significantly increased LaminB1 levels overall ($p = 0.016$, Fig. 3i; Table 2d). Follow-up analysis of condition-specific effects showed that NK92 co-culture significantly increased LaminB1 in A β -treated neurons ($p = 0.008$), but not in non-A β -treated neurons ($p = 0.456$, Fig. 3j; Table 2d). Nuclear HMGB1 analysis did not reach significance ($p = 0.101$, Table 2e). Likewise, condition-specific comparisons did not reveal significant differences between treatments (Fig. 3k; Table 2e). Our analysis for LaminB1 and HMGB1 could indicate underlying changes in neuron populations that simultaneously expressed low levels of LaminB1 and nuclear HMGB1 (HMGB1^{Lo}/LaminB1^{Lo}). To assess this, we identified HMGB1^{Lo}/LaminB1^{Lo} neurons and calculated the loss of these populations from

VEH to NK92 (VEH Δ NK92) and from A β to A β + NK92 (A β Δ A β + NK92), as described in Sup. Figure 3e. A Bonferroni correction for the repeated use of HMGB1 and LaminB1 datapoints across three tests established the threshold for significance at $p < 0.0167$. This correction was applied to control the familywise error rate, which increases with multiple testing. For example, finding a significant effect in LaminB1 alone increases the likelihood of a false-positive result when the same datapoints are reused in a combined analysis with HMGB1, thereby inflating the overall error rate. The analysis revealed that the loss of HMGB1^{Lo}/LaminB1^{Lo} neurons for A β Δ A β + NK92 was significantly higher than for VEH Δ NK92 ($p = 0.009$; Fig. 3l; Table 2f; Sup. Figure 3e, f). This is consistent with the elimination of senescent-like cells identified by both low LaminB1 and low nuclear HMGB1. We conducted a similar analysis for GFAP-positive astrocytes. The assessment of live astrocytes did not reveal significant differences in viability (Fig. 4a; Table 2g), indicating lack of effects of NK92 on astrocyte live cell counts as analyzed. However, a two-way ANOVA revealed that NK92 co-culture significantly increased overall nuclear aCasp3 in astrocytes regardless of A β treatment ($p = 0.009$; Fig. 4b; Table 2h) and double GFAP/aCasp3-positive astrocytes were readily identifiable (Sup. Figure 3c), indicating non-selective elimination was taking place to some degree. Condition-specific analysis of nuclear aCasp3 revealed a significant difference between

A β and A β + NK92 treatment groups ($p = 0.012$) that was not observed between VEH and NK92 groups ($p = 0.150$; Fig. 4c; Table 2h), suggesting preferential elimination of A β -treated astrocytes was taking place. However, the analysis of aCasp3 signal within GFAP-positive astrocytic processes showed NK92 co-culture did not yield significant results (Fig. 4d; Table 2i).

The analysis of senescence biomarkers did not yield significant differences for LaminB1 (Fig. 4e; Table 2j). Nuclear HMGB1 analysis did not reveal significant effects or trends in main effects or interactions (Fig. 4f; Table 2k). When combining LaminB1 and HMGB1 data and applying the same Bonferroni correction ($p < 0.0167$) as performed previously for neurons (Sup. Figure 3e), the analysis showed a highly significantly greater loss of HMGB1^{Lo}/LaminB1^{Lo} astrocytes in A β Δ A β + NK92 compared to VEH Δ NK92 ($p < 0.001$; Fig. 4g; Table 2l; Sup. Figure 4).

Our results reveal the preferential elimination of A β -treated neurons and astrocytes with senescent-like phenotypes, yet NK92 cells also exhibited cytotoxicity towards vehicle-control neurons and astrocytes. This effect was particularly pronounced in two co-cultures that were excluded from analysis due to severe cytotoxicity associated with NK92 cells. In these cases, VEH (Video 10) and A β (Video 11) control cultures showed abundant, healthy neurons and astrocytes, supporting the validity and proper execution of the experiments. In

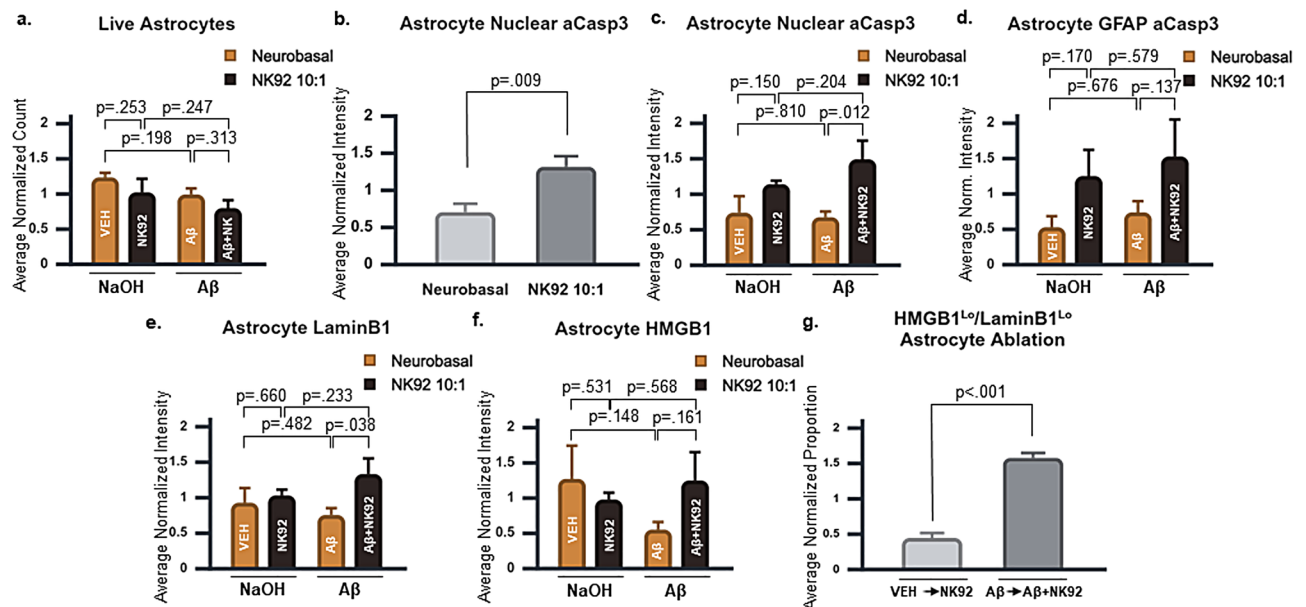


Fig. 4 In vitro AD model of senescence immune surveillance in human astrocytes. **(a)** A β x NK92 simple main effects live astrocyte count. **(b)** NK92 main and **(c)** A β x NK92 simple main effects on nuclear aCasp3. **(d, e, f)** A β x NK92 simple main effects of **(d)** aCasp3 signal intensity within GFAP positive processes, **(e)** LaminB1 and **(f)** HMGB1. **(g)** Comparison of HMGB1^{Lo}/LaminB1^{Lo} GFAP astrocytes between VEH Δ NK92 and A β Δ A β + NK92, as described in Sup Fig.3e. **(a, c, d, e, f)** Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. **(b)** Main effects. Significance, $p < 0.05$. **(g)** Independent sample *t*-test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. 3x Biological Replicates. A total of 1661 astrocytes were analyzed. Graphs represent mean $\pm$ s.e.m

contrast, co-cultures with NK92 cells led to widespread elimination of cells in both vehicle (Video 12) and A β -treated (Video 13) conditions. Because these replicates exhibited near total cell loss, it was not possible to accurately assess the selective elimination of a subpopulation of cells, prompting their exclusion from quantitative analysis. Nonetheless, these excluded replicates further support the conclusion that NK92 cells can induce non-selective cytotoxicity, including off-target killing of healthy cells. Altogether, our results are aligned with senescence immune surveillance of A β -treated cells in the context of non-selective cytotoxicity.

Navitoclax toxicity is not compatible with the Preferential elimination of senescent cells in A β in vitro model of AD

NAV inhibits the Bcl-2 family of anti-apoptotic proteins and is extensively used as a senolytic compound [18, 48–50]. It was originally reported to selectively kill mouse astrocytes and spare mouse neurons to achieve therapeutic effects in a transgenic mouse model of tauopathy [12]. To gain further insights into the effect of NAV on human neurons and astrocytes, cultures were treated with the vehicle for A β or A β 0.5 μ M at 28 DIV as performed previously for NK92 cell experiments (Fig. 3a). NAV was added at 14 days post-treatment (DPT) for 24 h and cells were fixed for ICC. For our initial experiments we chose NAV treatments at 4 and 8 μ M. The concentrations of NAV used in our study were selected based on our previous analysis of in vitro experiments evaluating NAV and other senolytics across a range of cell types, experimental conditions, and dose levels [18]. We intentionally chose concentrations from the higher end of the in vitro-tested range, as some cell types are known to be non-responsive to NAV [18]. By applying higher concentrations, we aimed to determine whether NAV exerted any effect on neurons or astrocytes, or whether it was innocuous in our system.

Regardless of whether cells were treated with A β or not, aCasp3 positive processes were evident (Sup. Figure 5a). Although we did not detect double positive aCasp3 and MAP2 cells, we did observe aCasp3 positive cell remnants with long processes resembling dendrites and axons. To gain further insight into the morphology of neuron-like apoptotic remnants, we created a new aCasp3 channel (aCasp3*) that enhanced aCasp3 signal in cells that had already undergone cell death as a function of their loss for the rest of the markers, as described in Sup. Figure 5b. This new channel revealed extensive aCasp3* positive processes that were consistent with axonal and dendritic morphology (Sup. Figure 5b, c). In addition to aCasp3, neuronal loss could also be inferred from the visually evident deterioration of MAP2 positive processes in NAV-treated cultures irrespective of A β treatment (Sup. Figure 5d). GFAP positive puncta also

appeared after NAV treatment but the effects on astrocyte viability were not as evident (Sup. Figure 5d).

We used the processed aCasp3* channel (Sup. Figure 5b) to determine whether there were differences in apoptotic cell death between treatment conditions. To quantify apoptotic cell death, we calculated the ratio of live cells to aCasp3*-positive cell remnants negative for MAP2 and GFAP, as described in Sup. Figure 5e. Of note, this analysis was not feasible for NK92 experiments, as there was no way to distinguish between the apoptotic cell remnants of neurons, astrocytes, and NK92 cells (Fig. 3b; Sup. Figure 3b, c). The analysis showed that NAV treatments significantly increased apoptotic cell death regardless of A β treatment ($p < 0.001$; Sup. Figure 5f; Table 3a), reflecting lack of selectivity. We also assessed differences in the volume of MAP2 and GFAP-positive processes per unit area, revealing a significant reduction in neuronal processes at all concentrations ($p < 0.001$; Sup. Figure 5g; Table 3b) and a significant reduction in astrocytic processes at 4 μ M NAV ($p = 0.038$; Sup. Figure 5h; Table 3c) relative to VEH that was independent of A β exposure, corroborating the lack of selectivity at these concentrations.

Thus, NAV did affect the viability of our cultures, ruling out the possibility that both human primary neurons and astrocytes in these cultures were impervious to NAV. However, concentrations of 4 and 8 μ M were likely too toxic to reliably assess the elimination of potentially small subpopulations of cells. We therefore tested NAV at 0.5 μ M. At this concentration, NAV has been shown to induce preferential elimination of senescent cells under various conditions and across multiple cell types [18]. The experimental design mirrored that employed in the NK92 and high-concentration NAV experiments, but with NAV at 0.5 μ M (Fig. 3a). MAP2 and GFAP negative apoptotic cell remnants could be identified in the original aCasp3 channel under NAV treatment irrespective of the presence of A β (Fig. 5a). Following the same approach used in the 4 μ M and 8 μ M NAV treatment experiments (Sup. Figure 5b), we generated an aCasp3* channel from the original aCasp3. As in higher-dose treatments, aCasp3* signal was also evident in NAV-treated conditions with or without A β (Sup. Figure 5i). However, unlike for higher NAV concentrations, the effects of NAV at 0.5 μ M on MAP2 and GFAP processes did not indicate obvious impacts on neuron and astrocyte populations (Sup. Figure 5j). Together, this suggested a moderate effect of NAV at 0.5 μ M.

For apoptotic cell death, a two-way ANOVA revealed that NAV at 0.5 μ M significantly increased apoptosis overall ($p = 0.031$; Fig. 5b; Table 4a). After applying a Bonferroni correction to adjust for two simple main effects ($p < 0.025$), further condition-specific analysis did not reveal significant differences (Fig. 5c; Table 4a). However,

Table 3 Navitoclax 4 & 8 μ M

Variable	Effects	Min Shapiro-Wilk	Levene's test	ANOVA/Bonferroni Multiple Comparisons
a Apoptotic Cell-Death See Sup. Fig.5e.	Interaction	0.531	0.395	$F(2, 12) = 0.699, p = 0.516$, partial $\eta^2 = 0.104$
	A β O			$F(1, 12) = 0.355, p = 0.563$, partial $\eta^2 = 0.029$
	NAV			$F(2, 12) = 18.738, p < 0.001$, partial $\eta^2 = 0.757$
	4 μ M Vs DMSO			Mean Diff 0.7434, 95% CI 0.1175 to 1.3694, $p = 0.019$
	8 μ M Vs DMSO			Mean Diff 1.3773, 95% CI 0.7513 to 2.0032, $p < 0.001$
	NAV 8 Vs 4 μ M			Mean Diff 0.6338, 95% CI 0.0079 to 1.2598, $p = 0.047$
b MAP2 Volume Average normalized ROI volume by scan area	Interaction	0.219	0.102	$F(2, 12) = 0.413, p = 0.670$, partial $\eta^2 = 0.064$
	A β O			$F(1, 12) = 1.081, p = 0.319$, partial $\eta^2 = 0.083$
	NAV			$F(2, 12) = 21.065, p < 0.001$, partial $\eta^2 = 0.778$
	4 μ M Vs DMSO			Mean Diff -0.9875, 95% CI -1.4636 to -0.5114, $p < 0.001$
	8 μ M Vs DMSO			Mean Diff -0.9360, 95% CI -1.4121 to -0.4600, $p < 0.001$
	NAV 8 Vs 4 μ M			Mean Diff 0.0515, 95% CI -0.4246 to 0.5275, $p = 1.000$
c GFAP Volume Average normalized ROI volume by scan area	Interaction	0.149	0.683	$F(2, 12) = 2.435, p = 0.130$, partial $\eta^2 = 0.289$
	A β O			$F(1, 12) = 0.457, p = 0.512$, partial $\eta^2 = 0.037$
	NAV			$F(2, 12) = 5.104, p = 0.025$, partial $\eta^2 = 0.460$
	4 μ M Vs DMSO			Mean Diff -0.3791, 95% CI -0.7383 to -0.198, $p = 0.038$
	8 μ M Vs DMSO			Mean Diff -0.0477, 95% CI -0.4070 to -0.3115, $p = 1.000$
	NAV 8 Vs 4 μ M			Mean Diff 0.3314, 95% CI -0.0279 to 0.6906, $p = 0.074$

no significant differences were detected in the viable neuron population remaining after NAV treatment (Fig. 5d; Table 4b), nuclear aCasp3 levels (Fig. 5e; Table 4c), or aCasp3 within MAP2-positive processes (Fig. 5f; Table 4d). These findings suggest that the apoptotic cell remnant may not have been neurons or the effects of NAV 0.5 μ M were mild enough to avoid detection by our metrics.

Surprisingly, NAV treatment significantly reduced the population mean of LaminB1 in neurons overall ($p = 0.047$; Fig. 5g; Table 4e), opposite to what would be expected from the elimination of a senescent-like neuron population characterized by low LaminB1 expression. Condition-specific analysis did not reveal additional insights (Fig. 5h; Table 4e), indicating reduction in LaminB1 was taking place irrespective of A β treatment. HMGB1 analysis did not yield significant effects or trends (Fig. 5i; Table 4f). As with NK92 cells, a Bonferroni correction for the repeated use of HMGB1 and LaminB1 required a $p < 0.0167$ to achieve significance. The assessment of HMGB1^{Lo}/LaminB1^{Lo} neurons showed no significant differences between VEH Δ NAV and A β Δ A β + NAV ($p = 0.058$; Fig. 5j; Table 4g; Sup. Figure 5 k). Thus, any undetected preferential killing associated with NAV treatment would not involve the selective elimination of HMGB1^{Lo}/LaminB1^{Lo} neurons.

In terms of the effects on astrocytes, there were no observable significant results. The analysis of live astrocytes (Fig. 6a; Table 4h), aCasp3 signal within astrocyte nuclei (Fig. 6b; Table 4i), and aCasp3 within GFAP positive processes (Fig. 6c; Table 4j) did not reveal significant effects. Similarly, the analysis of astrocytic LaminB1 (Fig. 6d; Table 4k), HMGB1 (Fig. 6e; Table 4l), and

HMGB1^{Lo}/LaminB1^{Lo} astrocytes (significance declared at $p < 0.0167$; Fig. 6f; Sup. Figure 6 ; Table 4m) did not provide statistical support for selective elimination of cells by NAV.

Overall, NAV treatment at high concentrations (4 and 8 μ M) resulted in non-selective elimination of both neurons and astrocytes. The analysis of apoptotic cell death following lower-concentration NAV treatment (0.5 μ M) further supported the lack of selectivity. Cell type-specific viability and apoptosis metrics, however, did not clearly indicate whether neurons or astrocytes were preferentially affected at this lower dose. As a result, we remain cautious about whether toxicity occurs in neurons at 0.5 μ M. Nonetheless, even if selective cell elimination was occurring undetected, NAV 0.5 μ M did not target cells in a way that is consistent with the elimination of cells expressing low LaminB1 or low nuclear HMGB1.

Senolytic cocktail treatment with DQ does not provide evidence of selective killing of senescent cells in A β in vitro model of AD

Dasatinib acts on tyrosine kinases, including Src family kinases and Bcr-Abl [51, 52]. In contrast, Quercetin interferes with anti-apoptotic pathways such as Bcl-2 and AKT survival signaling, among other targets [53, 54]. The combination of Dasatinib and Quercetin in the DQ formulation was strategically developed to leverage the senolytic properties of both compounds, enabling the targeted elimination of a broader spectrum of senescent cells at lower doses [87]. DQ has been reported to ameliorate disease in a mouse model of AD-like tau pathology by selectively killing mouse senescent-like neurons whilst sparing glial cells [11]. However, probably due to

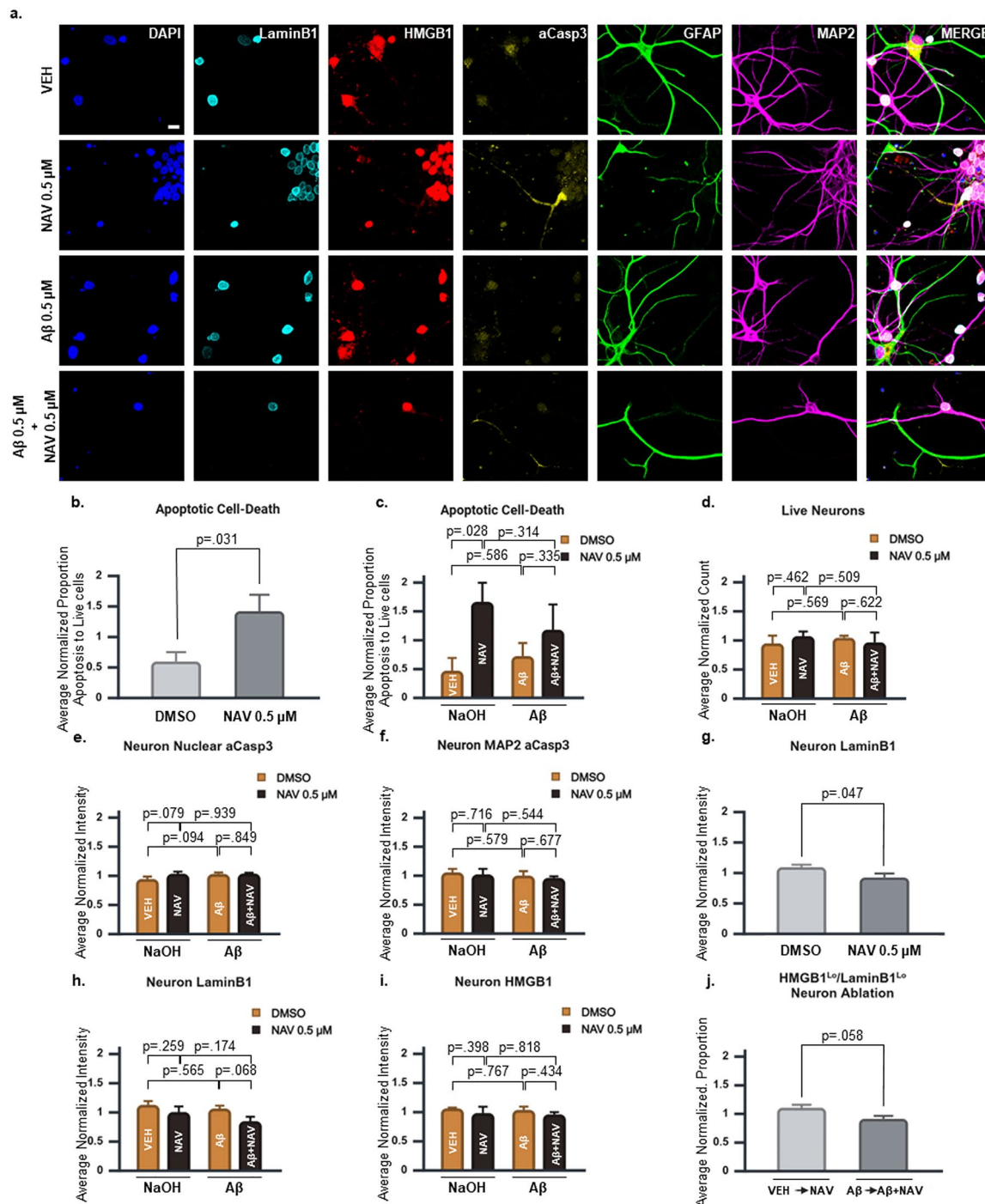


Fig. 5 Assessment of NAV 0.5 μ M on human primary neurons. **(a)** Representative linearly unmixed confocal spectral scans of all treatment conditions of NAV 0.5 μ M experiments. 40x objective with 1x zoom. Scale bar = 10 μ m. **(b)** NAV main effects and **(c)** A β x NAV simple main effects on apoptotic cell death, quantified as described in Sup. Figure 5e. **(d, e, f)** A β x NAV simple main effects on **(d)** live neuron count, **(e)** nuclear aCasp3 and **(f)** aCasp3 within MAP2 processes. **(g)** NAV main effects and **(h)** A β x NAV simple main effects on LaminB1. **(i)** A β x NAV simple main effects on HMGB1. **(j)** Comparison of HMGB1^{Lo}/LaminB1^{Lo} neuronal loss for VEH>NAV and A β >A β +NAV, as described in Sup. Fig. 3e. **(b, g)** Main effects. Significance, $p < 0.05$. **(c, d, e, f, h, i)** Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. **(j)** Independent sample t -test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. 3x Biological Replicates. A total of 9840 neurons were analyzed. Graphs represent mean $\pm$ s.e.m

Table 4 Navitoclax 0.5 μ M

Variable	Effects	Min Shapiro-Wilk	Levene's test	ANOVA/Kruskal-Wallis/t-test
a Apoptotic Cell-Death See Sup. Fig. 5e.	Interaction	0.118	0.554	$F(1, 8) = 1.347, p = 0.279$, partial $\eta^2 = 0.144$
	A β O			$F(1, 8) = 0.128, p = 0.730$, partial $\eta^2 = 0.016$
	NAV 0.5			$F(1, 8) = 6.825, p = 0.031$, partial $\eta^2 = 0.460$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 0.323, p = 0.586$, partial $\eta^2 = 0.039$
	NAV	NaOH Vs. A β O		$F(1, 8) = 1.152, p = 0.314$, partial $\eta^2 = 0.126$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 7.118, p = 0.028$, partial $\eta^2 = 0.471$
	A β O	DMSO Vs. NAV		$F(1, 8) = 1.054, p = 0.335$, partial $\eta^2 = 0.116$
b Neuron Viability: Average normalized total cell count	Interaction	0.197	0.256	$F(1, 8) = 0.826, p = 0.390$, partial $\eta^2 = 0.094$
	A β O			$F(1, 8) = 0.005, p = 0.946$, partial $\eta^2 = 0.001$
	NAV 0.5			$F(1, 8) = 0.033, p = 0.859$, partial $\eta^2 = 0.004$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 0.352, p = 0.569$, partial $\eta^2 = 0.042$
	NAV	NaOH Vs. A β O		$F(1, 8) = 0.478, p = 0.509$, partial $\eta^2 = 0.056$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 0.596, p = 0.462$, partial $\eta^2 = 0.069$
	A β O	DMSO Vs. NAV		$F(1, 8) = 0.263, p = 0.622$, partial $\eta^2 = 0.032$
c Neuron Nuclear aCasp3: Average normalized mean nuclear aCasp3 signal	Interaction	0.18	0.164	$F(1, 8) = 1.654, p = 0.234$, partial $\eta^2 = 0.171$
	A β O			$F(1, 8) = 1.955, p = 0.200$, partial $\eta^2 = 0.196$
	NAV 0.5			$F(1, 8) = 2.446, p = 0.156$, partial $\eta^2 = 0.234$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 3.602, p = 0.094$, partial $\eta^2 = 0.310$
	NAV	NaOH Vs. A β O		$F(1, 8) = 0.006, p = 0.939$, partial $\eta^2 = 0.001$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 4.061, p = 0.079$, partial $\eta^2 = 0.337$
	A β O	DMSO Vs. NAV		$F(1, 8) = 0.039, p = 0.849$, partial $\eta^2 = 0.005$
d aCasp3 within MAP2: Average normalized aCasp3 signal per unit volume of MAP2	Interaction	0.221	0.239	$F(1, 8) = 0.002, p = 0.970$, partial $\eta^2 = 0.000$
	A β O			$F(1, 8) = 0.735, p = 0.416$, partial $\eta^2 = 0.084$
	NAV 0.5			$F(1, 8) = 0.327, p = 0.583$, partial $\eta^2 = 0.039$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 0.335, p = 0.579$, partial $\eta^2 = 0.040$
	NAV	NaOH Vs. A β O		$F(1, 8) = 0.402, p = 0.544$, partial $\eta^2 = 0.048$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 0.142, p = 0.716$, partial $\eta^2 = 0.017$
	A β O	DMSO Vs. NAV		$F(1, 8) = 0.187, p = 0.677$, partial $\eta^2 = 0.023$
e Neuron LaminB1 Average normalized mean LaminB1 signal	Interaction	0.363	0.51	$F(1, 8) = 0.398, p = 0.546$, partial $\eta^2 = 0.047$
	A β O			$F(1, 8) = 2.192, p = 0.177$, partial $\eta^2 = 0.215$
	NAV 0.5			$F(1, 8) = 5.512, p = 0.047$, partial $\eta^2 = 0.408$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 0.361, p = 0.565$, partial $\eta^2 = 0.043$
	NAV	NaOH Vs. A β O		$F(1, 8) = 2.229, p = 0.174$, partial $\eta^2 = 0.218$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 1.474, p = 0.259$, partial $\eta^2 = 0.156$
	A β O	DMSO Vs. NAV		$F(1, 8) = 4.436, p = 0.068$, partial $\eta^2 = 0.357$
f Neuron nuclear HMGB1: Average normalized mean nuclear HMGB1 signal	Interaction	0.247	0.144	$F(1, 8) = 0.002, p = 0.962$, partial $\eta^2 = 0.000$
	A β O			$F(1, 8) = 0.149, p = 0.710$, partial $\eta^2 = 0.018$
	NAV 0.5			$F(1, 8) = 1.473, p = 0.259$, partial $\eta^2 = 0.156$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 0.094, p = 0.767$, partial $\eta^2 = 0.012$
	NAV	NaOH Vs. A β O		$F(1, 8) = 0.057, p = 0.818$, partial $\eta^2 = 0.007$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 0.797, p = 0.398$, partial $\eta^2 = 0.091$
	A β O	DMSO Vs. NAV		$F(1, 8) = 0.679, p = 0.434$, partial $\eta^2 = 0.078$
g Neuron HMGB1Lo/LaminB1Lo See Sup. Fig. 3e	NAV/A β O+NAV	0.717	1	$t(4) = 2.633, p = 0.058$
h Astrocyte Viability: Average normalized total cell count	Interaction	0.100	0.208	$F(1, 8) = 2.807, p = 0.132$, partial $\eta^2 = 0.260$
	A β O			$F(1, 8) = 0.395, p = 0.547$, partial $\eta^2 = 0.047$
	NAV 0.5			$F(1, 8) = 1.970, p = 0.198$, partial $\eta^2 = 0.198$
	DMSO	NaOH Vs. A β O		$F(1, 8) = 2.654, p = 0.142$, partial $\eta^2 = 0.249$
	NAV	NaOH Vs. A β O		$F(1, 8) = 0.548, p = 0.480$, partial $\eta^2 = 0.064$
	NaOH	DMSO Vs. NAV		$F(1, 8) = 4.739, p = 0.061$, partial $\eta^2 = 0.372$
	A β O	DMSO Vs. NAV		$F(1, 8) = 0.037, p = 0.852$, partial $\eta^2 = 0.005$
i Astrocyte Nuclear	Interaction	0.136	0.320	$F(1, 8) = 1.533, p = 0.251$, partial $\eta^2 = 0.161$

Table 4 (continued)

Variable	Effects	Min Shapiro-Wilk	Levene's test	ANOVA/Kruskal-Wallis/t-test
aCasp3:	A β O			F(1, 8) = 1.055, p = 0.334, partial η^2 = 0.117
Average normalized	NAV 0.5			F(1, 8) = 3.166, p = 0.113, partial η^2 = 0.284
mean nuclear	DMSO	NaOH Vs. A β O		F(1, 8) = 2.566, p = 0.148, partial η^2 = 0.243
aCasp3 signal	NAV	NaOH Vs. A β O		F(1, 8) = 0.022, p = 0.885, partial η^2 = 0.003
	NaOH	DMSO Vs. NAV		F(1, 8) = 4.553, p = 0.065, partial η^2 = 0.363
	A β O	DMSO Vs. NAV		F(1, 8) = 0.146, p = 0.712, partial η^2 = 0.018
j aCasp3 within GFAP:	Interaction	0.102	0.619	F(1, 8) = 0.640, p = 0.447, partial η^2 = 0.074
Average normalized	A β O			F(1, 8) = 0.024, p = 0.880, partial η^2 = 0.003
aCasp3 signal per unit	NAV 0.5			F(1, 8) = 3.780, p = 0.088, partial η^2 = 0.321
volume of GFAP	DMSO	NaOH Vs. A β O		F(1, 8) = 0.457, p = 0.518, partial η^2 = 0.054
	NAV	NaOH Vs. A β O		F(1, 8) = 0.207, p = 0.661, partial η^2 = 0.025
	NaOH	DMSO Vs. NAV		F(1, 8) = 3.765, p = 0.088, partial η^2 = 0.320
	A β O	DMSO Vs. NAV		F(1, 8) = 0.655, p = 0.442, partial η^2 = 0.076
k Astrocyte LaminB1	Interaction	0.419	0.097	F(1, 8) = 0.189, p = 0.676, partial η^2 = 0.023
Average normalized	A β O			F(1, 8) = 0.099, p = 0.761, partial η^2 = 0.012
mean LaminB1 signal	NAV 0.5			F(1, 8) = 2.080, p = 0.187, partial η^2 = 0.206
	DMSO	NaOH Vs. A β O		F(1, 8) = 0.007, p = 0.934, partial η^2 = 0.001
	NAV	NaOH Vs. A β O		F(1, 8) = 0.280, p = 0.611, partial η^2 = 0.034
	NaOH	DMSO Vs. NAV		F(1, 8) = 0.508, p = 0.496, partial η^2 = 0.060
	A β O	DMSO Vs. NAV		F(1, 8) = 1.761, p = 0.221, partial η^2 = 0.180
l Astrocyte nuclear HMGB1:	VEH/NAV/A β O/A β O+NAV	0.010		X ² (3) = 2.590, p = 0.459
Ranked Average normalized mean nuclear HMGB1 signal				
m Astrocyte HMGB1Lo/LaminB1Lo	NAV/A β O+NAV	0.82	1	t(4) = 1.319, p = 0.257
See Sup. Fig. 3e				

the use of different transgenic mouse models, others did not report senescent neuron elimination by DQ [16].

As with NAV, DQ does not eliminate all senescent cell types [18]. For this reason, as with NAV, we also began with higher concentrations to determine whether these had an effect. Experiments with Dasatinib (4 μ M) and Quercetin (30 μ M) followed the same methodology as those conducted for NK92 and NAV (Fig. 3a). Visual inspection did not reveal obvious pro-apoptotic effects of DQ at these doses, either in the original aCasp3 channel (Fig. 7a) or after signal enhancement in the aCasp3* channel (Sup. Figure 7a). Similarly, MAP2 and GFAP signals remained unaffected (Sup. Figure 7b).

A non-parametric analysis revealed no statistically significant differences between treatments on apoptotic cell death (p = 0.319; Fig. 7b; Table 5a). There were also no statistically significant differences in the number of viable neurons analyzed across treatment conditions (Fig. 7c; Table 5b). For the analysis of nuclear aCasp3 in neurons, condition-specific analysis only revealed significantly increased nuclear aCasp3 signal in A β -treated neurons relative to VEH (p = 0.022; Fig. 7d; Table 5c). For somato-dendritic aCasp3, a non-parametric Kruskal-Wallis

H test did not reveal significant differences (p = 0.218; Fig. 7e; Table 5d).

For analysis of cellular senescence biomarkers, LaminB1 analysis showed no significant differences that would be consistent with the elimination of specific neuron populations (Fig. 7f; Table 5e). For HMGB1, subgroup analysis did not reveal significant differences or trends indicative of a biologically relevant senolytic effect (Fig. 7g). Likewise, applying the correction for the repeated use of the dependent variables (p < 0.0167), no significant changes were observed in HMGB1^{Lo}/LaminB1^{Lo} neurons when comparing the effects of DQ treatment in healthy vs. A β -treated neurons, providing no evidence of selective elimination (p = 0.444; Fig. 7h; Table 5g; Sup. Figure 7 c).

As per astrocytes, a reduction in the number of analyzed cells was associated with DQ treatment irrespective of A β exposure (p = 0.022; Fig. 8a; Table 5h), indicating non-selective loss of viability. Condition-specific analysis did not reveal significant results (Fig. 8b; Table 5h). For nuclear aCasp3, there were no significant effects (Fig. 8c; Table 5i). Likewise, a non-parametric ANOVA assessing aCasp3 within GFAP-positive processes did not detect statistically significant differences (p = 0.347; Fig. 8d;

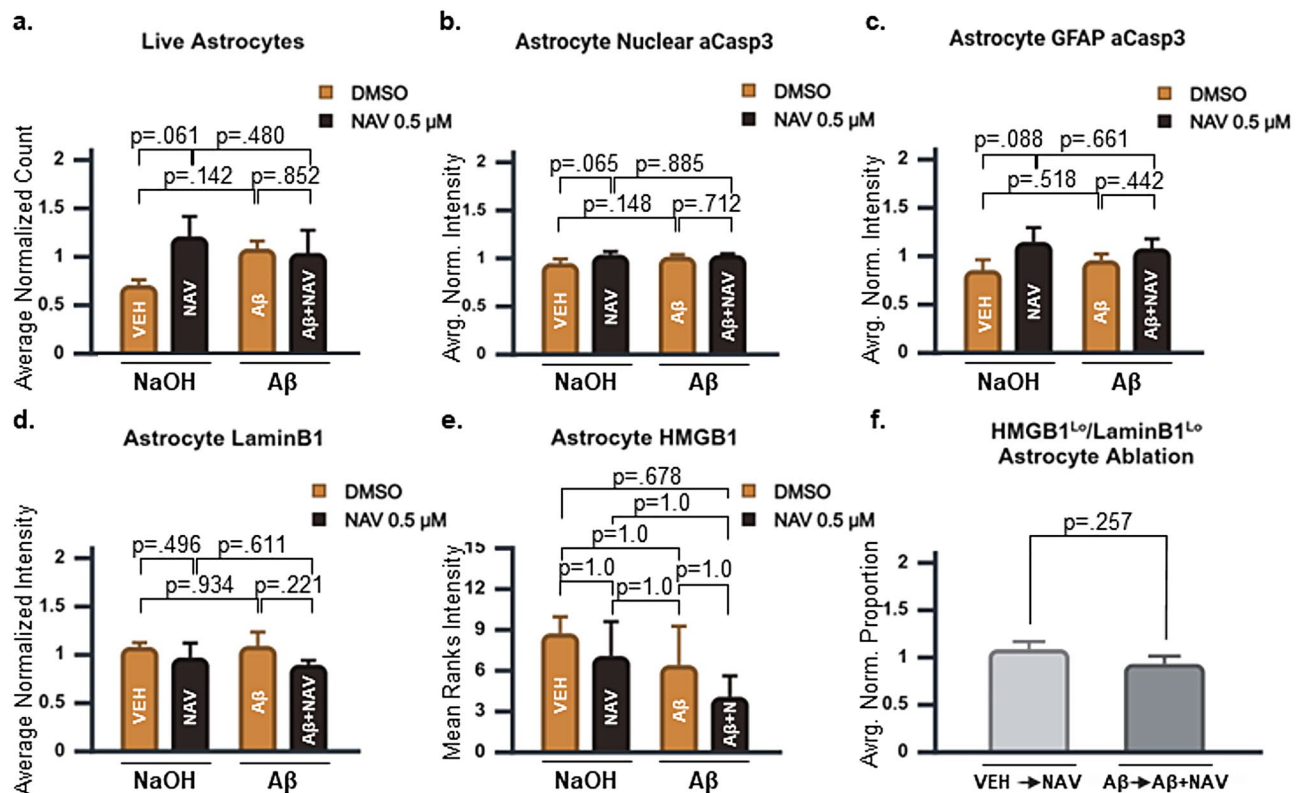


Fig. 6 Assessment of NAV 0.5 μ M on human primary astrocytes. (a, b, c, d) A β x NAV simple main effects on (a) live astrocyte count, (b) nuclear aCasp3, (c) aCasp3 within GFAP processes, and (d) LaminB1 signal. (e) Mean ranks for nuclear HMGB1. (f) Comparison of HMGB1^{Lo}/LaminB1^{Lo} astrocyte loss for VEH $\rightarrow$ NAV and A β $\rightarrow$ A β +NAV, as described in Sup. Fig. 3e. (a, b, c, d) Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. (e) Bonferroni-corrected all pairwise comparisons between ranks. Significance, $p < 0.05$. (f) Independent sample t -test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. 3x Biological Replicates. A total of 2216 astrocytes were analyzed. Graphs represent mean $\pm$ s.e.m

Table 5j). The analysis of LaminB1 (Fig. 8e; Table 5k), nuclear HMGB1 (Fig. 8f; Table 5l), and HMGB1^{Lo}/LaminB1^{Lo} astrocytes (significance declared at $p < 0.0167$; Fig. 8g; Sup. Figure 8a; Table 5m) also did not show significant differences, arguing against selective killing of astrocytes by DQ.

A β -induced neurotoxicity observed in the DQ treatment condition (Fig. 7d) was not seen in the NK92 (Fig. 3f) or NAV (Fig. 5e) experiments, which may reflect variability in A β treatment effects. However, direct comparisons across compounds with different patterns of selectivity must be interpreted cautiously when using average-normalized dependent variables (Sup. Figure 8 b). Further, some variability in A β toxicity is expected given the duration of the treatments. While rat primary neuron cultures exhibit robust A β -induced toxicity within 24 hours [88, 89] 24-hour exposure to 10 μ M extracellular A β (1–42) has been reported to have no significant effect on human neuronal viability [27]. High-density human primary neuron cultures typically require prolonged exposure—10 to 14 days—to 1–20 μ g/mL A β (1–40) (approximately 0.23–4.62 μ M) to induce overt neuronal death [40]. Nevertheless, in the case A β toxicity

was indeed more pronounced in the DQ experiment, this would create favorable conditions to observe selective senolytic activity. However, the absence of DQ effects on neurons and astrocytes rules out senolytic activity as assessed, as well as non-selective toxicity to neurons. While our analysis suggests that DQ may induce mild, non-selective toxicity in astrocytes, statistically significant results were inconsistent across metrics, indicating that any effect—if present—is likely less pronounced than that observed with NK92 co-culture.

Discussion

We describe a straightforward method for culturing primary human neurons and astrocytes from commercially available cryopreserved cell preparations. These cultures can be maintained well beyond 100 DIV, with GFAP-negative neurons expressing NeuN, MAP2, tau, and Ankyrin-G. By 28–43 DIV, the expression of synaptic maturation markers establishes a well-defined timeline for modeling CNS diseases with adult-onset characteristics.

We briefly describe several unexpected phenotypes observed in our very long-term human primary neuron cultures, including protracted viability, NeuN staining

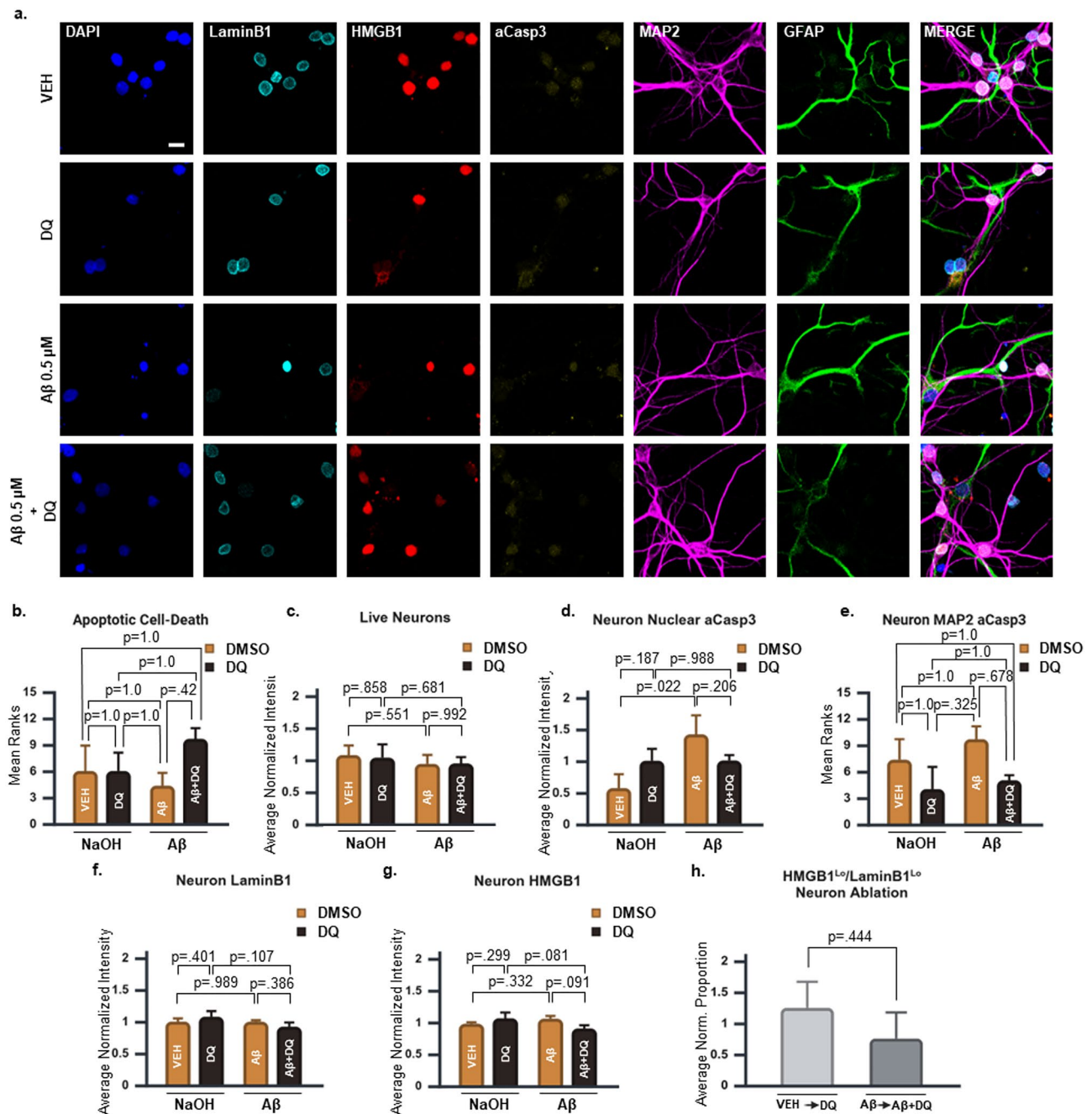


Fig. 7 Assessment of Dasatinib 4 μM + Quercetin 30 μM senolytic cocktail on human primary neurons. **(a)** Representative linearly unmixed confocal spectral scans of all treatment conditions of DQ experiments. 40x objective with 1x zoom. Scale bar = 10 μm. **(b)** Mean ranks of apoptotic cell death in Aβ x DQ treatment groups, as quantified in Sup. Fig. 5e. **(c, d)** Aβ x DQ simple main effects on **(c)** live neuron count and **(d)** nuclear aCasp3. **(e)** Mean ranks of aCasp3 signal within MAP2 processes in Aβ x DQ treatment groups. **(f, g)** Aβ x DQ simple main effects on **(f)** LaminB1 and **(g)** nuclear HMGB1 signal. **(h)** Comparison of HMGB1^{Lo}/LaminB1^{Lo} neuronal loss for VEH>DQ and Aβ>Aβ+DQ, as quantified in Sup. Fig. 3e. **(b, e)** Bonferroni-corrected all pairwise comparisons between ranks. Significance, $p < 0.05$. **(c, d, f, g)** Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. **(h)** Independent sample t-test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. A total of 12313 neurons were analyzed. 3x Biological replicates. Graphs represent mean $\pm$ s.e.m.

extending into processes, nuclear MN423 immunoreactivity, and faint axonal MC-1 staining. The extended viability of human compared to murine primary neurons might be unsurprising given the pronounced lifespan

differences between species. Nevertheless, these findings could reflect a combination of factors, including protracted culturing-induced phenotypic drift, previously uncharacterized features of human neurons,

Table 5 Dasatinib (4 μ M) and Quercetin (30 μ M)

Variable	Effects	Min Shapiro-Wilk	Lev- ene's test	ANOVA/Kruskal-Wallis/t-test
a Apoptotic Cell-Death See Sup. Fig. 5e.	VEH/DQ/A β O/A β O+DQ	0.010		X2(3) = 3.513, p = 0.319
b Neuron Viability: Average normalized total cell count	Interaction A β O DQ DMSO DQ NaOH A β O NaOH Vs. A β O NaOH Vs. A β O DMSO Vs. DQ DMSO Vs. DQ	0.173	0.506	F(1, 8) = 0.019, p = 0.894, partial η^2 = 0.002 F(1, 8) = 0.551, p = 0.479, partial η^2 = 0.064 F(1, 8) = 0.015, p = 0.905, partial η^2 = 0.002 F(1, 8) = 0.387, p = 0.551, partial η^2 = 0.046 F(1, 8) = 0.182, p = 0.681, partial η^2 = 0.022 F(1, 8) = 0.034, p = 0.858, partial η^2 = 0.004 F(1, 8) = 0.000, p = 0.992, partial η^2 = 0.000
c Neuron Nuclear aCasp3: Average normalized mean nuclear aCasp3 signal	Interaction A β O DQ DMSO DQ NaOH A β O NaOH Vs. A β O NaOH Vs. A β O DMSO Vs. DQ DMSO Vs. DQ	0.58	0.299	F(1, 8) = 3.969, p = 0.082, partial η^2 = 0.332 F(1, 8) = 4.058, p = 0.079, partial η^2 = 0.337 F(1, 8) = 0.002, p = 0.963, partial η^2 = 0.000 F(1, 8) = 8.026, p = 0.022, partial η^2 = 0.501 F(1, 8) = 0.000, p = 0.988, partial η^2 = 0.000 F(1, 8) = 2.080, p = 0.187, partial η^2 = 0.206 F(1, 8) = 1.891, p = 0.206, partial η^2 = 0.191
d aCasp3 within MAP2: Ranked average norm. aCasp3 signal per unit volume of MAP2	VEH/DQ/A β O/A β O+DQ (1 Extreme Outlier)	0.095		X2(3) = 4.436, p = 0.218
e Neuron LaminB1 Average normalized mean LaminB1 signal	Interaction A β O DQ DMSO DQ NaOH A β O NaOH Vs. A β O NaOH Vs. A β O DMSO Vs. DQ DMSO Vs. DQ	0.306	0.357	F(1, 8) = 1.626, p = 0.238, partial η^2 = 0.169 F(1, 8) = 1.676, p = 0.232, partial η^2 = 0.173 F(1, 8) = 0.000, p = 0.983, partial η^2 = 0.000 F(1, 8) = 0.000, p = 0.989, partial η^2 = 0.000 F(1, 8) = 3.302, p = 0.107, partial η^2 = 0.292 F(1, 8) = 0.786, p = 0.401, partial η^2 = 0.089 F(1, 8) = 0.841, p = 0.386, partial η^2 = 0.095
f Neuron nuclear HMGB1: Average normalized mean nuclear HMGB1 signal	Interaction A β O DQ DMSO DQ NaOH A β O NaOH Vs. A β O NaOH Vs. A β O DMSO Vs. DQ DMSO Vs. DQ	0.403	0.408	F(1, 8) = 4.596, p = 0.064, partial η^2 = 0.365 F(1, 8) = 0.467, p = 0.514, partial η^2 = 0.055 F(1, 8) = 0.330, p = 0.582, partial η^2 = 0.040 F(1, 8) = 1.066, p = 0.332, partial η^2 = 0.118 F(1, 8) = 3.996, p = 0.081, partial η^2 = 0.333 F(1, 8) = 1.232, p = 0.299, partial η^2 = 0.133 F(1, 8) = 3.693, p = 0.091, partial η^2 = 0.316
g Neuron	DQ/A β O+DQ	0.998	1	t(4) = 0.848, p = 0.444
HMGB1Lo/LaminB1Lo See Sup. Fig. 3e				
h Astrocyte Viability: Average normalized total cell count	Interaction A β O DQ DMSO DQ NaOH A β O NaOH Vs. A β O NaOH Vs. A β O DMSO Vs. DQ DMSO Vs. DQ	0.186	0.572	F(1, 8) = 0.600, p = 0.461, partial η^2 = 0.070 F(1, 8) = 3.158, p = 0.113, partial η^2 = 0.283 F(1, 8) = 8.045, p = 0.022, partial η^2 = 0.501 F(1, 8) = 3.255, p = 0.109, partial η^2 = 0.289 F(1, 8) = 0.503, p = 0.499, partial η^2 = 0.059 F(1, 8) = 6.519, p = 0.034, partial η^2 = 0.449 F(1, 8) = 2.216, p = 0.183, partial η^2 = 0.210
i Astrocyte Nuclear aCasp3: Average normalized mean nuclear aCasp3 signal	Interaction A β O DQ DMSO DQ NaOH Vs. A β O NaOH Vs. A β O	0.064	0.061	F(1, 8) = 0.156, p = 0.703, partial η^2 = 0.019 F(1, 8) = 1.571, p = 0.245, partial η^2 = 0.164 F(1, 8) = 3.959, p = 0.082, partial η^2 = 0.331 F(1, 8) = 0.368, p = 0.561, partial η^2 = 0.044 F(1, 8) = 1.359, p = 0.277, partial η^2 = 0.145

Table 5 (continued)

Variable	Effects	Min Shapiro-Wilk	Levene's test	ANOVA/Kruskal-Wallis/t-test
j aCasp3 within GFAP: Ranked average norm. aCasp3 signal per unit volume of GFAP	NaOH	0.065		$F(1, 8) = 1.271, p = 0.292$, partial $\eta^2 = 0.137$
	A β O			$F(1, 8) = 2.844, p = 0.130$, partial $\eta^2 = 0.262$
	VEH/DQ/A β O/A β O+DQ (1 Extreme Outlier)			$X^2(3) = 3.308, p = 0.347$
k Astrocyte LaminB1 Average normalized mean LaminB1 signal	Interaction	0.401	0.419	$F(1, 8) = 1.096, p = 0.326$, partial $\eta^2 = 0.121$
	A β O			$F(1, 8) = 0.711, p = 0.424$, partial $\eta^2 = 0.082$
	DQ			$F(1, 8) = 0.266, p = 0.620$, partial $\eta^2 = 0.032$
	DMSO			$F(1, 8) = 0.021, p = 0.889$, partial $\eta^2 = 0.003$
	DQ			$F(1, 8) = 1.786, p = 0.218$, partial $\eta^2 = 0.183$
	NaOH			$F(1, 8) = 0.141, p = 0.717$, partial $\eta^2 = 0.017$
	A β O			$F(1, 8) = 1.222, p = 0.301$, partial $\eta^2 = 0.132$
l Astrocyte nuclear HMGB1: Ranked Average normalized mean nuclear HMGB1 signal	Interaction	0.215	0.804	$F(1, 8) = 0.441, p = 0.525$, partial $\eta^2 = 0.052$
	A β O			$F(1, 8) = 0.460, p = 0.517$, partial $\eta^2 = 0.054$
	DQ			$F(1, 8) = 2.690, p = 0.140$, partial $\eta^2 = 0.252$
	DMSO			$F(1, 8) = 0.000, p = 0.992$, partial $\eta^2 = 0.000$
	DQ			$F(1, 8) = 0.901, p = 0.370$, partial $\eta^2 = 0.101$
	NaOH			$F(1, 8) = 0.476, p = 0.510$, partial $\eta^2 = 0.056$
	A β O			$F(1, 8) = 2.655, p = 0.142$, partial $\eta^2 = 0.249$
m Astrocyte HMGB1Lo/LaminB1Lo See Sup. Fig. 3e	VEH/DQ/A β O/A β O+DQ	0.657	1	$t(4) = 1.224, p = 0.288$

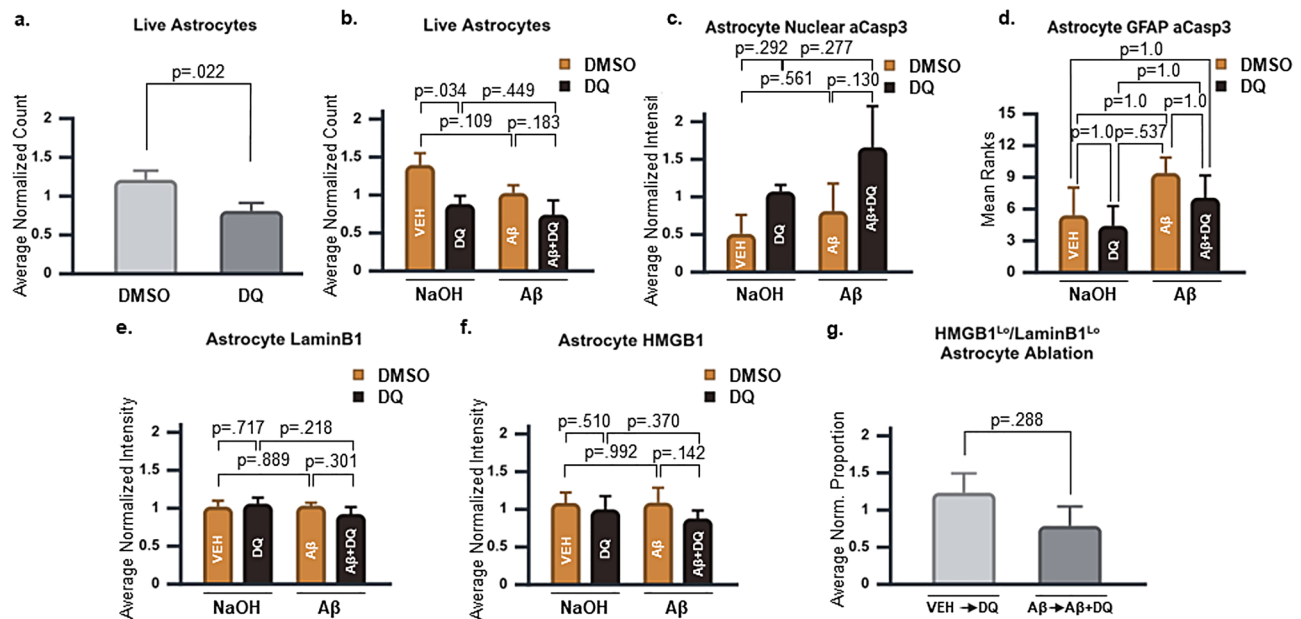


Fig. 8 Assessment of Dasatinib 4 μ M + Quercetin 30 μ M senolytic cocktail on human primary astrocytes. **(a)** DQ main effects and **(b)** A β x DQ simple main effects on live astrocyte count. **(c)** A β x DQ simple main effects on nuclear aCasp3. **(d)** Mean ranks of aCasp3 signal within GFAP processes in A β x DQ treatment groups. **(e, f)** A β x DQ simple main effects on **(e)** LaminB1, and **(f)** nuclear HMGB1 signal. **(g)** Comparison of HMGB1Lo/LaminB1Lo astrocyte loss for VEH>DQ and A β >A β +DQ, as quantified in Sup. Fig. 3e. **(a, d)** Bonferroni-corrected all pairwise comparisons between ranks. Significance, $p < 0.05$. **(b, c, e, f)** Simple main effects with Bonferroni correction for two simple main effects. Significance, $p < 0.025$. **(g)** Independent sample t-test with Bonferroni correction for 3 omnibus tests, two-tailed. Significance, $p < 0.0167$. 3x Biological replicates. A total of 2658 astrocytes were analyzed. Graphs represent mean $\pm$ s.e.m.

or methodological challenges inherent to studying the human brain—some of which may be more readily addressed *in vitro*. For instance, both MC-1 and Alz-50 target pathological tau conformers, yet Alz-50 is known to cross-react with fetal Alz-50 reactive clone 1 (FAC1), a nuclear protein predominantly expressed in the adult human cortex [90–92]. Similarly, our MN423 nuclear staining could indicate previously unrecognized antibody cross-reactivity. Additionally, faint axonal Alz-50 immunoreactivity in healthy rat and human primary neurons has been previously reported [26, 34, 93] which may explain faint MC-1 positive axons. Taken together, these observations raise intriguing possibilities but will require further investigation to clarify their biological significance and distinguish between true phenotypes, technical artifacts, and antibody cross-reactivity.

NAV, DQ, and NK92 cells: insights into senolytic activity and selectivity

We utilized these cultures to model AD pathology and investigate senolytic selectivity as a proof of principle for their potential application in AD therapeutic testing.

To the best of our knowledge, the BBB permeability of NAV in humans has not been established. In contrast, recent Phase I clinical trial data in AD indicate that Dasatinib, but not Quercetin, is detectable in the cerebrospinal fluid (CSF) [94] suggesting limited BBB permeability for the DQ cocktail. The brain is also an immune-privileged site, with restricted access of peripheral immune cells to the parenchyma [95]. Nonetheless, the mechanisms of action of NAV, DQ, and NK cells are well characterized [18, 55] making them valuable tools for probing senolytic strategies and for guiding the development of novel compounds with improved BBB permeability for CNS applications. The cell culture system described here lacks a BBB, thereby enabling direct exposure of primary human neurons and astrocytes to NAV, DQ, and NK92 cells and facilitating such mechanistic assessment.

NK92 co-culture was the only regimen in which the metrics for viability and the analysis of putative senescence markers LaminB1 and HMGB1 collectively supported the targeted elimination of senescent-like neurons and astrocytes treated with A β . However, elimination was partially selective at best, based on ample evidence of toxicity to neurons and astrocytes in the absence of A β . This aligns with previous findings on primary NK cytotoxicity on astrocytes [96]. The off-target cytotoxicity observed in this study raises concerns that CNS cells may exhibit cross-reactivity with emerging immune-based senolytic therapies designed to leverage NK signaling for senescent cell clearance [97, 98]. Further studies are required to assess this potential risk.

Previous studies in mice and non-human primates have not reported NAV as neurotoxic [12, 16, 99] but our results demonstrate that NAV can kill human primary neurons in a concentration-dependent manner *in vitro*. As noted, to the best of our knowledge, the BBB permeability of NAV has not been characterized in humans. Consequently, it remains unknown whether NAV, when administered for non-CNS indications, can reach the concentrations we tested *in vitro* within the human brain parenchyma and induce toxic side effects. Nonetheless, our findings highlight a potential neurotoxicity risk that may not have been evidenced *in vivo* in murine models.

Noteworthy, off-target toxicity of NAV in non-senescent cell types is well documented across multiple *in vitro* studies [18]. From a mechanistic standpoint, such toxicity is not inherently problematic *in vitro*—provided NAV selectively eliminates a substantially greater proportion of senescent cells compared to non-senescent cells. In that case, a theoretical dose should exist at which only senescent cells are targeted. In contrast, a non-selective compound would exhibit similar toxicity to senescent and non-senescent cells regardless of concentration, with reductions in dose yielding proportionally reduced effects but no increase in selectivity. This is precisely what we observe with NAV, despite testing concentrations that probe the range in which NAV is reported selective in other cells [18].

Thus, even if NAV is ultimately safe for peripheral applications due to limited BBB permeability, our findings raise concerns about targeting CNS cells with BBB-permeable agents that rely on the same mechanism of action. NAV acts by inhibiting the Bcl-2 family of anti-apoptotic proteins [18, 48–50] and our results therefore suggest that indiscriminate targeting of this family may not be viable for CNS interventions. Future research should focus on identifying specific Bcl-2 family members that can be targeted more selectively to preserve senolytic efficacy while minimizing off-target effects. This type of refinement is not unprecedented. NAV's original application in oncology revealed dose-limiting thrombocytopenia caused by inhibition of Bcl-XL, a Bcl-2 family member essential for platelet survival [100–102]. This led to the development of Venetoclax, a more selective Bcl-2 inhibitor designed to avoid Bcl-XL-mediated toxicity [103]. Based on our results, similar precision may be required for NAV-inspired CNS-directed senolytic therapies.

In contrast, DQ did not exhibit overt neurotoxicity, even at higher concentrations commonly used *in vitro* [18]. While non-selective astrocyte toxicity was suggested by total cell counts, aCasp3-based viability metrics were inconsistent, indicating that any observed effect was likely moderate. No clear senolytic activity was detected based on LaminB1 or HMGB1, diverging from previous

reports in transgenic mouse models where DQ selectively eliminated senescent-like neurons or OPCs [11, 13]. However, the emergence of deleterious senescent-like neurons in previous mouse studies was attributed to tau pathology [11] which may not be present in our cultures. Although previous studies on mouse OPCs used mouse models of A β pathology comparable to our in vitro experiments [13] we neither studied nor assessed the presence of OPCs in our cultures. Consequently, our results with DQ are consistent with prior findings in mice. Importantly, a Phase I clinical trial in AD reported detectable levels of dasatinib, but not quercetin, in the CSF, and deemed the intervention safe for human use [94]. Given quercetin's CSF levels, the high concentrations tested in our study are unlikely to be clinically achievable, which in our view supports the safety profile of the aforementioned phase I clinical trials.

While experiments with NK92 co-culture were consistent with the elimination of senescent cell populations and NAV experiments were not, neither regimen is arguably safe to target senescent cell populations that represent a small percentage of the total cell population. For example, a realistic scenario based on published in vitro data involves 76% on-target killing, derived from experiments in which the senolytic compound was tested on a pure population of senescent cells in culture, and 25% off-target toxicity, measured in a separate culture composed exclusively of non-senescent cells [18]. This yields an apparent selectivity of 51% (76% – 25%), which may initially seem promising. However, such a level of selectivity would have catastrophic implications in mixed cell populations, as would be encountered in vivo. To illustrate, if senescent cells comprise only 1% of the total population, the same 76% on-target and 25% off-target killing would result in 76 senescent cells eliminated and 2,475 non-senescent (healthy) cells killed in a population of 10,000 cells. This equates to over 3,250 healthy cells lost for every 100 senescent cells eliminated, a tradeoff that would be unacceptable in clinical settings, especially in non-regenerative tissues such as the brain. Even if NK92 cells exhibit preferential cytotoxicity, our findings suggest their risk profile is comparable to the lack of selectivity seen with NAV at high concentrations. Future studies incorporating tau pathology in our model should determine whether DQ selectively eliminates senescent cells as reported in mice [11].

Limitations

Our work is not intended to be the final word on the mechanisms of action of NK92 cells, NAV, or DQ, but rather to underscore that the translatability of findings from mouse models to human systems cannot be taken for granted. Each senolytic mechanism examined in this study warrants further detailed investigation to validate

our findings and to fully elucidate the nature and specificity of their effects. For instance, incorporating additional markers of apoptotic and non-apoptotic cell death (e.g., Annexin V, propidium iodide) could offer deeper insight into both the mode and extent of cytotoxicity. Moreover, the use of small interfering RNA (siRNA) to selectively knock down proteins targeted by these senolytics could help distinguish which survival pathways are essential for healthy neurons and astrocytes—and thus may need to be preserved in human interventions [50]. While our findings provide valuable insights into the effects of NAV, DQ, and NK92 cells, they must be interpreted cautiously due to several limitations. First, mixed primary neuron and astrocyte cultures do not fully recapitulate the complexity of the brain, which includes additional cell types and a highly organized structural architecture. The mixed human primary neuron and astrocyte cultures described here are not intended to replace other models but rather to expand the available toolkit by offering a human-relevant, reductionist system amenable to mechanistic investigation. Regarding cellular senescence, we did not detect a distinct senescent-like population when comparing control and A β -treated cells. A β induces a wide range of cellular and molecular toxicities unrelated to cellular senescence [104] which may have obscured small senescent-like populations that were revealed only under NK92 co-culture. Future studies should incorporate additional senescence markers and techniques such as single cell RNA sequencing while acknowledging the known challenges of identifying senescent neurons [78, 79]. Another limitation is that NK cells eliminate both “stressed” and senescent cells through NKG2D receptor activation [55, 105]. Since A β is a known cellular stressor [104] NK92 cells may have targeted stressed cells rather than or in addition to senescent cells. Additionally, NK92 cells possess a simplified receptor repertoire [106] and our cultures lack additional immune cells required to regulate NK cytotoxicity in vivo. Future studies should assess NK cytotoxicity in a more physiologically relevant context. Finally, astrocyte assessments may be confounded by the use of GFAP, a marker linked to astrocyte reactivity rather than astrocyte identity [107] emphasizing the need for additional markers to facilitate a more refined analysis.

Conclusions

We developed an accessible method for culturing primary human neurons and astrocytes to maturity using commercially available cryopreserved cell preparations. This system is straightforward to establish, requiring only basic cell culture skills, and can be readily adapted for studying a range of CNS models and therapeutic interventions. The robustness of our human in vitro AD model is demonstrated by its ability to differentiate the effects

ANTIBODIES					
CELL CULTURE CHARACTERIZATION					
Antibody	Host/Type	Supplier	Catalog #	Dilution	Feature
3R tau (8E6/C11)	Mx mono. IgG	Sigma	05-803	1:100	Axon
Ankyrin-G (N106/36)	Mx mono. IgG2a	Sigma	MABN466	1:50	Axon Initial Segment
GFAP	Rb poly.	Abcam	ab7620	1:1000	Astrocytes
GFAP (2.2B10)	Rat mono. IgG2a	Invitrogen	13-0300	1:1500	Astrocytes
Homer-1	Mx mono. IgG1	SYSY	160 011	1:200	Postsynaptic
MAP2	Ch poly.	Abcam	ab5392	1:2500	Neurons
MC1 anti-tau	Rb poly.	Absolute Antibody	Ab02389-23.0	1:100	Axon
MN423 anti-tau	Mx mono. IgG1	Abcam	ab80579	1:50	Axon
NeuN (1B7)	Mx mono. IgG2b	Abcam	ab104224	1:500-1:1000	Neurons
PSD-95 (6G6-1C9)	Mx mono. IgG2a	ThermoFisher	MA1-045	1:500	Postsynaptic
S100 β (SH-B1)	Mx mono. IgG1	Sigma	S2532	1:1000	Astrocytes
Synapsin-1 (A6442)	Rb Ppoly.	ThermoFisher	A6442	1:250	Presynaptic
AMYLOID BETA ANTIBODIES					
Antibody	Host/Type	Supplier	Catalog #	Dilution	Feature
4G8 anti-A β	Mx mono.	Biolegend	800709	1:4000	A β /APP
6E10 anti-	Mx mono. IgG1	Biolegend	803001	1:500	A β /APP
OC anti-A β	Rb poly.	Sigma	AB2286	1:4000	Fibrillar A β
SENOLYTIC EXPERIMENTS ANTIBODIES					
Antibody	Host/Type	Supplier	Catalog #	Dilution	Feature
aCasp-3 (5A1E)	Rb poly.	Cell Signaling	9664	1:250	Apoptosis
GFAP	Rb poly.	Abcam	ab7620	1:1000	Astrocytes
HMGB1 (1F3)	Mx mono. IgG2a	Abcam	ab190377	1:1000	Nuclei
Lamin B1	Rb poly.	Abcam	ab16048	1:1000	Nuclear Lamina
MAP2	Ch poly.	Abcam	ab5392	1:2500	Neurons
SECONDARY ANTIBODIES					
Antibody	Goat Host/Type	Supplier	Catalog #	Dilution	Feature
Alexa Fluor 488	Mx IgG1	Invitrogen	A-21121	1:1000	Secondary
Alexa Fluor 488	Mx IgG	Invitrogen	A-11001	1:1000-1:2000	Secondary
Alexa Fluor 488	Rb IgG	Invitrogen	A-11008	1:500	Secondary
Alexa Fluor 546	Mx IgG	Invitrogen	A-11030	1:500	Secondary
Alexa Fluor 546	Mx IgG2b	Invitrogen	A-21143	1:1000	Secondary
Alexa Fluor 546	Rb IgG	Invitrogen	A-11035	1:500	Secondary
Alexa Fluor 594	Mx IgG1	Invitrogen	A-11030	1:500	Secondary
Alexa Fluor 594	Rb IgG	Invitrogen	A-32740	1:1000	Secondary
Alexa Fluor 633	Rat IgG	Invitrogen	A-21094	1:1000	Secondary
Alexa Fluor 647	Ch IgY	Invitrogen	A-21449	1:1000-1:2000	Secondary
Alexa Fluor 647 Plus	Rb IgG	Invitrogen	A-32733	1:500	Secondary

Table 6 Antibodies

of NK92 cells, NAV, and DQ, aligning with their distinct mechanisms of action [18]. These findings emphasize the importance of human-specific models in refining therapeutic strategies for neurodegenerative diseases prior to their advancement to clinical applications.

Materials and methods

Antibodies

A complete list of the antibodies and their dilutions can be found in Table 6.

Mixed human primary neuron and astrocyte cultures

To prepare Ibidi “ibiTreat” 8-well chamber slides (Ibidi, 80826) for neuronal cultures, each well was coated with 200 μ L of a 1:1 solution of 1x Dulbecco's phosphate-buffered saline (VWR, VWRL0119-0500) and Poly-D-Lysine (PDL; ThermoFisher, A3890401). The slides were placed in Petri dishes, serving as secondary containers to minimize contamination risk, and incubated overnight at 37 °C in a sterile humidified atmosphere. The following day, the wells were washed by aspirating the DPBS/PDL solution and immediately adding 300 μ L of 1x DPBS to prevent drying. This washing process was repeated twice, waiting at least 2 h between washes. The culture media was prepared before cell plating and consisted of Neurobasal Plus Medium (ThermoFisher, A3582901), 50x B-27 Plus Supplement (ThermoFisher, A3582801), 100x GlutaMAX Supplement (ThermoFisher, 35050061), and 100x Penicillin-Streptomycin solution (ThermoFisher, 15070063). Filter sterilization of the solution with a 0.22 μ m syringe filter (VWR, 76479-024) prior to adding the B-27 Plus Supplement had no observable effects on cell viability. Before plating, the wells of the slides were washed with pre-warmed culture media and 240 μ L of media was added to each well to accommodate approximately a 60 μ L cell suspension, ensuring a final volume around 300 μ L. The prepared slides were equilibrated in a humidified incubator at 37 °C with 5% CO₂ and 3–5% O₂ for at least 30 min before cell plating. O₂ levels at 3–5% were used to mimic physiological oxygen levels for cellular senescence experiments, but the cells may be cultured at the more conventional 20%. Vials with human fetal brain tissue between 17 and 19 gestational weeks were provided by Sciencell (Sciencell, 1520). Each cryovial ($\sim 1 \times 10^6$ cells) was briefly opened/closed to release pressure and thawed rapidly in a 37 °C water bath for approximately 2 min until a small ice crystal remained. The vial was then cleaned with ethanol and the cell suspension was gently tapped to dislodge cells adhering to the cap. A 50 mL centrifuge tube pre-washed with plating media was prepared to prevent cell adhesion to the tube walls. Similarly, pipette tips were primed with media before aspirating cell suspensions to minimize cell loss during handling. The cells were carefully transferred to the pre-washed tube using primed tips, with the vial cap also aspirated to recover any remaining cells. One milliliter of pre-warmed media was added dropwise to the suspension to avoid osmotic shock, followed by the gentle addition of 2 mL of media to achieve a final concentration of $\sim 2.5 \times 10^5$ cells/mL. The cell suspension was gently pipetted to dissociate clumps and cell density confirmed using a hemocytometer. Cells were plated onto the equilibrated Ibidi slides at a density of $\sim 1\text{--}1.5 \times 10^4$ cells/cm², resulting in a total well volume of 300 μ L and placed into humidified incubator at 37 °C with 5% CO₂ and 3–5% O₂.

Completely replacing the media after plating to remove residual cryopreservants significantly reduced cell viability. Omitting the media replacement step and leaving the cells completely undisturbed for 48 h post-plating substantially improved culture health. Subsequent media changes were performed every 2 days whenever feasible, replacing only one-third of the media with fresh culture media lacking Penicillin-Streptomycin. Retaining sufficient conditioned media during these changes was essential to support the cells. Adjustments to the media volume were made to account for evaporation, ensuring the total well volume remained consistent with the initial plating volume. At the specified plating density, anti-mitotics were not required as astrocyte growth did not adversely affect neuronal viability or the overall health of the cultures. While some slight variations in cell density and volume can occur, both the overall cell-to-volume and cell-to-area ratios described herein were found to be important for the long-term viability of the cultures.

NK92 culture and co-culture

NK92 cells (ATCC, CRL-2407) were cultured following the vendor's instructions. Complete growth media was prepared using alpha MEM media (Gibco, 12561072) supplemented with 12.5% non-heat-inactivated fetal bovine serum (VWR, 45000-73), 12.5% heat-inactivated horse serum (ThermoFisher, 26050088), 0.02 mM folic acid (Sigma, F8758-5G), 0.2 mM myo-Inositol (Sigma, I7508-50G), 0.1 mM beta-mercaptoethanol (ThermoFisher, 21985023), 100 U/mL Penicillin-Streptomycin solution (ThermoFisher, 15070063), and 100 U/mL human recombinant IL-2 (R&D Systems, 202-IL). The prepared media was equilibrated in T25 flasks (Corning) in humidified incubator at 37 °C with 5% CO₂ and 20% O₂. Thawed NK92 cells were centrifuged at 800 x g for 5 min, resuspended in pre-warmed complete growth media, and transferred to pre-equilibrated T25 flasks at an initial density of $2\text{--}4 \times 10^5$ cells/mL. Cultures were maintained in a humidified incubator and split every 2–3 days, with the culture medium refreshed to sustain optimal cell density and viability. For co-culture experiments, NK92 cells were harvested, pelleted at 340 x g for 7 min, and resuspended in neurobasal medium. The cells were subsequently added to 42 DIV mixed primary human neuron and astrocyte cultures at an effector-to-target (E:T) ratio of 10:1 and co-cultured for 24 h (Figs. 3 and 4) or 20 h (Videos 10–13).

Live imaging

Cells were cultured in Ibidi 8-well chamber slides with polymer coverslip bottoms coated with PDL. Live-cell imaging was conducted using a Zeiss LSM 780 confocal microscope (Carl Zeiss Microscopy GmbH, Jena, Germany) with 20x, 40x, or 63x objectives. Imaging was

performed in a sealed chamber maintained at 37 °C and 5% CO₂. Images and videos were captured using ZEISS ZEN Microscopy Software (RRID: SCR_013672). Details regarding objectives, time intervals between frames, and experiment durations are provided in the video legends.

Cytoskeleton visualization

The cytoskeleton was visualized by transducing Tubulin tagged with red fluorescent protein (Tubulin-RFP) using the CellLight Tubulin-RFP, BacMam 2.0 reagent (ThermoFisher, C10503), following the manufacturer's instructions. Briefly, the CellLight reagent was added to wells containing 15,000 cells at a multiplicity of infection of 70 particles per cell and incubated for 16 h. Subsequently, cells were observed under a 40x objective on the Zeiss LSM 780 confocal microscope.

A β & senolytic treatments

A β peptide preparations enriched in oligomers were generated using a Beta Amyloid (1–42) aggregation kit (rPeptide, A-1170-2). Lyophilized A β peptides were reconstituted in either 5 mM Tris or 10 mM NaOH at a concentration of 1 mg/mL, then diluted with HPLC-grade water and Tris-buffered saline (TBS) to create a 100 μ M stock solution. To enrich for oligomers, the stock solution was incubated at 37 °C for 3 h, aliquoted, and stored at -20 °C until use. Primary human neuronal-astrocytic cultures were treated with A β preparations or their vehicle at 28 DIV at a final concentration of 0.5 μ M. The senolytics Navitoclax (Selleckchem, S1001), Dasatinib (Selleckchem, S1021), and Quercetin (Selleckchem, S2391) were prepared according to the vendor's instructions. NAV (0.5, 4, or 8 μ M), Dasatinib (4 μ M) plus Quercetin (30 μ M), or a DMSO vehicle solution (final DMSO concentration <0.1%), were added to the cultures at 42 DIV. After 24 h of treatment, the cultures were collected for ICC.

Immunocytochemistry (ICC) and image acquisition

ICC and confocal imaging were performed on the microscopy-compatible Ibidi slides to ensure minimal disruption to cellular morphology. Cell cultures were fixed by progressively replacing the culture media with 4% paraformaldehyde (PFA; Santa Cruz, 3052589-4) and incubated with PFA at room temperature (RT) for 15 min, followed by three washes with ice-cold PBS for 10 min each. Permeabilization was performed using PBS containing 0.05% Triton x-100 (Northwest Scientific, JTB/x198-07) (PBTx) for 30 min. After permeabilization, cultures were incubated for 30 min at RT in PBTx supplemented with 10% normal goat serum (NGS; Jackson ImmunoResearch, 005-000-121) to block nonspecific antibody binding. Primary antibodies were diluted in PBS containing 1% NGS and applied to the cultures for

1 h at RT. This was followed by four 10-minute washes with PBS to remove unbound primary antibodies. Secondary antibodies, diluted in PBS with 1% NGS, were applied for 1 h at RT, followed by four additional 10-minute washes with PBS. For ICC involving repeated use of isotype-specific antibodies, a sequential staining protocol was employed. The first round of primary and secondary antibody incubations was performed on the first day. Following an overnight wash to remove residual antibodies, a second round of primary and secondary antibody incubations was carried out on the next day. DNA labeling was performed by incubating the cultures in PBS containing 1 μ g/mL 4',6-diamidino-2-phenylindole (DAPI; Invitrogen, D1306) for 3 min. After three final washes with PBS, the PBS was entirely replaced with pure glycerol (Invitrogen, 15514-011) to preserve the stained cells and minimize refractive index changes during imaging. Confocal imaging was performed using a Zeiss LSM 980 for cell culture characterization and Zeiss LSM 780 confocal microscopes (Carl Zeiss Microscopy GmbH, Jena, Germany) for cell culture characterization and senolytic regimen studies. The Zeiss LSM 980 was equipped with an Airyscan 2 detector, which enables super-resolution imaging through its 32-channel GaAsP detector array.

Spectral scanning with linear unmixing

To resolve DAPI and Alexa Fluor-conjugated secondary antibodies labeled with 488, 546, 594, 633, and 647 fluorophores, spectral imaging with linear unmixing was performed using the Zeiss LSM 780 confocal microscope. Imaging was conducted using the full spectral range of the 32-channel GaAsP spectral detector, along with the 405, 488, 561, and 633 nm excitation lasers. Spectral reference fingerprints for each primary–secondary antibody pair were acquired from single-stained primary neuron cultures, rather than from generic fluorophore databases, to accurately capture emission profiles within the specific background autofluorescence of the neuronal culture system. These reference spectra were applied in the system's online fingerprinting mode to enable real-time spectral unmixing during acquisition. Because all channels share a common gain setting, primary antibody clones were carefully selected and paired with secondary antibodies of similar expected fluorescence intensities. This approach optimized signal detection across the full fluorophore panel while avoiding saturation or under-detection caused by gain imbalances. To further optimize acquisition, primary antibodies targeting non-overlapping epitopes and exhibiting comparable signal intensities were assigned to the same excitation laser—for example, GFAP and MAP2 were labeled with Alexa Fluor 633 and Alexa Fluor 647, respectively, both excited by the 633 nm laser. This ensured consistent excitation efficiency and signal comparability under shared laser

power and gain conditions. For antibody targets expected to exhibit spatial overlap, fluorophores with maximally separated emission spectra were used to reduce spectral bleed-through and background autofluorescence. While spectral unmixing corrects for emission overlap computationally, this additional step ensured that each pixel within the lambda stack retained a clean and distinct emission profile, thereby improving the accuracy and fidelity of linear unmixing.

Image generation and processing

Maximum intensity projection (MIP) images and time-lapse videos were created using ZEISS ZEN Microscopy Software (RRID: SCR_013672). IMARIS software (RRID: SCR_007370) was used for 3D visualization, video generation, and processing of 3D confocal images. The algorithms used to generate the ROIs were optimized for each replicate and standardized across treatments within the same replicate to eliminate user bias.

Synapse characterization

ROI for Synapsin, Homer-1, and PSD95 were defined using the spot function in IMARIS. The sum voxel intensity within all ROIs for each DIV and replicate in each channel was calculated and used for statistical analysis of total protein levels. To quantify colocalization between Synapsin-Homer-1 and Synapsin-PSD95, spots generated for the above-described analysis were used to generate new channels with the synaptic markers. The IMARIS colocalization function was applied to generate a new channel containing colocalized voxels of the masked channels. The spot function was then used on the channel to generate ROI on colocalized puncta. The total number of ROIs for colocalized puncta was calculated for each DIV and replicate and used for statistical analysis.

Viability, LaminB1, and HMGB1 analysis for NK92, NAV 0.5 μ M, and DQ experiments

Nuclear ROIs were defined using the surface function applied to DAPI-stained nuclei. User-defined algorithms were utilized to classify cells as live or dead based on nuclear characteristics. Additional user-defined and machine learning algorithms were employed to further classify live cell nuclei ROI as belonging to neurons or astrocytes, with manual inspection performed to validate the classifications. The processed nuclear ROI data were exported as CSV files and imported into FlowJo (RRID: SCR_008520) for further refinement. Within FlowJo, neurons were identified as MAP2^{hi}/GFAP^{lo} and astrocytes as MAP2^{lo}/GFAP^{hi} by plotting MAP2 against GFAP channels. All analyzed neurons were live cells positive for MAP2 and negative for GFAP while all analyzed astrocytes were live cells negative for MAP2 and positive for GFAP. These populations were then exported into new

CSV files for subsequent statistical analyses. The total number of ROIs classified as neurons or astrocytes was used to analyze the viability of live cells. Mean HMGB1 and aCasp3 intensity values within the ROIs were used for statistical analyses of nuclear HMGB1 and aCasp3 levels. Additional steps were required to accurately assess LaminB1. Nuclear-based ROIs inherently overcapture the LaminB1 signal, a nuclear lamina protein, and average its signal with non-nuclear lamina regions within the ROI. This results in an underestimation of LaminB1 signal intensity in large nuclei relative to smaller nuclei. To address this issue, a new LaminB1 channel was created by setting LaminB1-positive voxels in the original channel to the maximum intensity (255 in an 8-bit image). Within each nuclear ROI, the sum intensity of this newly generated channel was divided by 255 to calculate the number of LaminB1-positive voxels, effectively excluding contributions from non-nuclear lamina regions. The sum signal intensity of the original LaminB1 channel was then divided by the number of LaminB1-positive voxels to obtain the true LaminB1 mean intensity for each ROI. This corrected LaminB1 mean was subsequently used for statistical analysis.

Apoptotic cell death analysis for NAV and DQ experiments

Pyknotic nuclei were identified in the DAPI channel using user-defined algorithms, generating corresponding ROIs. An enhanced aCasp3 channel (aCasp3*) was created by increasing the relative intensity of aCasp3-positive voxels inversely proportional to signals in other channels (as shown in Sup. Figure 5b). Noise in the aCasp3* channel was minimized using Gaussian, median, and threshold cutoff filters in IMARIS where necessary, standardizing the steps for all treatments within each replicate. This refined aCasp3* channel was used to define ROIs representing apoptotic cell remnants. These ROIs were size limited to fragment larger apoptotic remnants into smaller, discrete ROIs overlapping with pyknotic nuclei ROIs. The pyknotic and aCasp3* ROIs corresponding to the same apoptotic cell remnant were identified based on spatial proximity and visually inspected to ensure accuracy. Subsequently, these ROIs were used to mask a new channel from which final ROIs representing apoptotic cell bodies were generated (Sup. Figure 5e). Additional ROIs were created to identify live cells, irrespective of cell type, to determine the total number of live cells in the cultures. The proportion of apoptotic cell ROIs to live cell ROIs was calculated and used for statistical analysis, providing a measure of apoptotic cell death relative to overall cell viability within the cultures.

aCasp3 analysis in neuronal processes for NK92, NAV 0.5 μ M, and DQ experiments

ROI for neuronal and astrocytic processes were defined using the surface function of IMARIS on MAP2 and GFAP channels, respectively. The sum pixel intensity of aCasp3 within each ROI was calculated and divided by the ROI volume to obtain a volume-normalized aCasp3 signal. This normalized signal was used for statistical analyses to assess cytoplasmic aCasp3.

Volume analysis of neuronal and astrocytic processes for NAV 4 and 8 μ M experiments

To analyze neuronal and astrocytic processes, ROIs were created based on MAP2 and GFAP channels. The total volume of MAP2-positive or GFAP-positive ROIs was calculated and divided by the confocal image area to determine the volume of neuronal or astrocytic processes normalized to the image area, which was subsequently used for statistical analyses.

Statistics

Statistical analyses were conducted using IBM SPSS Statistics (RRID: SCR_002865). For an overview of the statistical workflow used in this study, see Sup. Figure 9a. For a conceptual explanation of two-way ANOVAs with interaction terms, see Sup. Figure 9b, c. Significant results are reported in the main text, while non-significant findings are omitted from the text for clarity. Instead, details of all statistical tests, including assumption tests, are presented in Tables 1, 2, 3, 4 and 5. Assumption tests were carried out to detect extreme outliers (box plots), normality (Shapiro-Wilk test, $p > 0.05$, and Q-Q plots), and homogeneity of variances (Levene's test, $p > 0.05$). Transformations appropriate to the type of assumption violation were attempted and assumption tests re-run. When transformed data did not meet the assumptions, non-parametric tests were run.

Synapse maturation

One-way ANOVA or Welch's F omnibus tests and Tukey's or Games-Howell multiple comparisons were performed as determined by Levene's test for homogeneity of variances. Statistical significance was set at $p < 0.05$ (two-tailed). The sum voxel signal intensity within the ROIs for Synapsin, Homer-1, and PSD95 was calculated replicate by replicate for each DIV to quantify total protein levels. Colocalization between Synapsin and Homer-1, as well as Synapsin and PSD95, was assessed by calculating the total number of ROIs capturing colocalized puncta, as defined in the image processing protocol. The analysis was performed on the log10 transform on the sum signal intensity or sum number of colocalized puncta for each DIV replicate by replicate.

Senolytic regimens

For the analysis of the effects of senolytic regimens on viable neurons, LaminB1, and nuclear HMGB1 measured by ROI based on nuclei, Mahalanobis distances were used to detect and discard multivariate outliers from the nuclei ROI data, with statistical significance assessed using a Chi-squared test ($p < 0.01$). The nuclear ROI predictive variables included voxel number, the length across the vertical axis, LaminB1 mean, HMGB1 mean, the shortest, mid, and longest distances within the ROI bounding box. After removing outliers, treatment means for cell count (Live neurons/astrocytes), nuclear aCasp3, LaminB1, and HMGB1 were calculated replicate by replicate and normalized to the average of each replicate. HMGB1^{Lo}/LaminB1^{Lo} were calculated as described in Sup. Figure 3e and normalized to the average of each replicate. For the analysis of apoptotic cell death (Sup. Figure 5e; see image processing), MAP2/GFAP volume per area (see image processing), and aCasp3 signal within MAP2/GFAP (see image processing), treatment means were calculated replicate by replicate and normalized to the average of each replicate. For the resulting variables, assumptions tests and transformations were carried out as described above for synapse maturation with the addition of scatter plots of predicted and residual values to complement Levene's test. A two-way ANOVA was used to assess the interaction and the main effects. For 2 $\times$ 2 ANOVA used in NK92, NAV 0.5 μ M, and DQ, the interaction assesses whether the difference between the double vehicle and the senolytic regimen is significantly different from the difference between A β treatments and A β plus the senolytic regimen. Regardless of the presence of an interaction, hypothesis-driven simple main effects for A β and for the senolytic regimen were run to identify differences relevant to the study while omitting irrelevant comparisons. The simple main effects for A β compare the effects of adding A β in the absence of senolytics and, separately, in the presence of senolytics. The simple main effects for the senolytic regimen compare the effects of adding senolytic regimen in the absence of A β and, separately, in the presence of A β . Bar graphs with simple main effect p values are always plotted. Main effects are only plotted when they reach statistical significance in the absence of a significant interaction. Bonferroni correction was applied to correct for two simple main effects, declaring significance at $p < 0.025$. When two-way ANOVA assumptions were not met, a non-parametric one-way ANOVA on ranks (Kruskal-Wallis) was run followed by Bonferroni all-pairwise comparisons ($p < 0.05$). The assessment of HMGB1^{Lo}/LaminB1^{Lo} populations was performed using two-tailed independent t-tests. Because the HMGB1 and LaminB1 datasets were previously analyzed in two-way ANOVAs, their reuse introduced the potential for inflated Type I error. To mitigate

this, a Bonferroni correction was applied for three tests, adjusting the significance threshold to $p < 0.0167$. This stricter threshold accounted for multiple analyses involving the same data to control for the likelihood of false-positive results. At least 3 biological replicates were used for the experiments, where each replicated corresponded to a different donor.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13195-025-01822-7>.

Supplementary Material 1: Video 1: Transmitted light time-lapse video of cells 2 hours after plating. Note lamellipodium actively probing the environment. 63x objective with 1.5x zoom.

Supplementary Material 2: Video 2: Confocal time-lapse video of cells expressing CellLight™ Tubulin-RFP (BacMam 2.0) for visualization of the cytoskeleton. 24 hours after plating. Note RFP-positive cells actively probing the environment. 40x objective with 0.6x zoom.

Supplementary Material 3: Video 3: Confocal time-lapse video of cells expressing CellLight™ Tubulin-RFP (BacMam 2.0) for visualization of the cytoskeleton. 24 hours after plating. Note a cell migrating from the cluster of cells (top left), clearly visible at time 06:26:19.331. Note cells undergoing cytokinesis (bottom right), starting at time 02:56:57.874. 40x objective with 0.6x zoom.

Supplementary Material 4: Video 4: Confocal plus transmitted light time-lapse video of cells expressing CellLight™ Tubulin-RFP (BacMam 2.0) for visualization of the cytoskeleton. 24 hours after plating. Note cytokinesis onset of central RFP-positive cell first detected at time 04:02:22.261. 40x objective with 0.6x zoom.

Supplementary Material 5: Video 5: Confocal plus transmitted light time-lapse video of cells expressing CellLight™ Tubulin-RFP (BacMam 2.0) for visualization of the cytoskeleton. 24 hours after plating. Note the astrocyte-like morphology of the RFP-positive cell (bottom, left corner) compared to the long axon-like projection of a potential developing neuron (top, right corner). 40x objective with 0.6x zoom.

Supplementary Material 6: Video 6: Transmitted light time-lapse video of control primary neuron and astrocyte cultures from 42 to 43 DIV. 40x objective with 1x zoom.

Supplementary Material 7: Video 7: Transmitted light time-lapse video of primary neuron and astrocyte cultures co-cultured with NK92 cells from 42 to 43 DIV. 40x objective with 1x zoom.

Supplementary Material 8: Video 8: Transmitted light time-lapse video of Aβ-treated primary neuron and astrocyte cultures co-cultured with NK92 cells from 42 to 43 DIV. Aβ treatments (0.5 μM) were carried out at 28 DIV. 40x objective with 1x zoom.

Supplementary Material 9: Video 9: Video of 3D reconstructed confocal spectral scan with linear unmixing images of 43 DIV mixed human primary neuron and astrocytes co-cultured with NK92 cells stained for MAP2 (violet) to identify neurons, GFAP (green) to identify astrocytes, LaminB1 (light green) and HMGB1 (not shown) as proxies for cellular senescence, DAPI (blue) for nuclei, and aCasp3 (yellow) to identify apoptotic cells. Close up of a double GFAP and aCasp3 positive astrocyte is shown at second 48.

Supplementary Material 10: Video 10: Video of 3D reconstructed confocal spectral scan with linear unmixing images of VEH-condition 43 DIV mixed human primary neuron and astrocyte cultures stained for MAP2 (violet) to identify neurons, GFAP (green) to identify astrocytes, DAPI (blue) for nuclei, and aCasp3 (yellow) to identify apoptotic cells. Neuronal (MAP2, violet) and astrocyte (GFAP, green) processes appear normal with few aCasp3 positive apoptotic cell remnants.

Supplementary Material 11: Video 11: Video of 3D reconstructed confocal spectral scan with linear unmixing images of 43 DIV mixed human

primary neuron and astrocyte cultures treated with Aβ and stained for MAP2 (violet) to identify neurons, GFAP (green) to identify astrocytes, DAPI (blue) for nuclei, and aCasp3 (yellow) to identify apoptotic cells. Neuronal (MAP2, violet) and astrocyte (GFAP, green) processes appear normal with few aCasp3 positive apoptotic cell remnants.

Supplementary Material 12: Video 12: Video of 3D reconstructed confocal spectral scan with linear unmixing images of 43 DIV mixed human primary neuron and astrocyte co-culture with NK92 cells stained for MAP2 (violet) to identify neurons, GFAP (green) to identify astrocytes, DAPI (blue) for nuclei, and aCasp3 (yellow) to identify apoptotic cells. Neuronal (MAP2, violet) and astrocyte (GFAP, green) processes strongly deteriorated with abundant aCasp3 positive apoptotic cell remnants and astrocytes, showcasing extensive off-target cytotoxicity to vehicle control cells.

Supplementary Material 13: Video 13: Video of 3D reconstructed confocal spectral scan with linear unmixing images of 43 DIV mixed human primary neuron and astrocyte treated with Aβ and co-cultured with NK92 cells stained for MAP2 (violet) to identify neurons, GFAP (green) to identify astrocytes, DAPI (blue) for nuclei, and aCasp3 (yellow) to identify apoptotic cells. Neuronal (MAP2, violet) and astrocyte (GFAP, green) processes strongly deteriorated with abundant aCasp3 positive apoptotic cell remnants and astrocytes, showcasing extensive cytotoxicity.

Supplementary Material 14: Supplementary Figures 1–9

Acknowledgements

We thank Tori Lazerson, Varun Sridhar, and Alexandra Steinberg for their work on cell cultures. We thank Allison Bischoff for insightful discussions. The figures were prepared using BioRender.

Author contributions

CCW, EW, SL, and CJS performed the experiments. Transductions were performed by EW and SL. Immunocytochemistry and confocal imaging were done by CCW, EW, SL, CJS, and ZM. Image processing was performed by CCW, EW, SL, NJB, and ZM. CCW performed statistical analysis. JKA and CCW supervised and guided the project. CCW wrote the manuscript, with input from all other authors.

Funding

This work is supported by the Larry L. Hillblom Foundation and the SENS research foundation.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethical approval

This research utilized de-identified primary neurons commercially obtained from ScienCell. ScienCell acquires human tissues from non-profit tissue providers who certify that all specimens are procured in accordance with local, state, and federal regulations governing the ethical collection and distribution of human biological materials.

Competing interests

The authors declare no competing interests.

Received: 14 February 2025 / Accepted: 16 July 2025

Published online: 26 July 2025

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